

Beyond Variance: Asymmetric Labor Income Risk and Portfolio Entry*

Tai Lo Yeung[†]

September 2026

Abstract

Variance compresses favorable and adverse earnings realizations into one scale measure. I use public CPS earnings histories and strictly later SIPP portfolios to separate lower-tail distance, upper-tail distance, total span, and orientation. Among 27,752 annual transitions beginning with no reported stock-and-mutual-fund holdings, a one-standard-deviation wider lower tail predicts a 2.71-percentage-point lower probability of positive holdings one year later and a wider upper tail a 1.92-point higher probability; the pair remains jointly distinguishable from zero under a 200-draw CPS first-stage sensitivity. With 99,999 cell-level wild-cluster draws, the lower and upper coefficients have two-component adjusted q -values of 0.019 and 0.087, and the signed composition contrast has $p = 0.033$. An exact rotation leaves total span imprecise and composition negative. Recomputing the CPS tails after excluding destination records classified as self-employed sharply attenuates the lower estimate. Across alternative zero-endpoint constructions, composition retains its direction and conditional precision; the standalone upper estimate remains positive but is imprecise in one construction. Historical group distributions are assigned rather than elicited, so the estimates are descriptive.

Keywords: labor income risk; skewness; measurement error; household portfolios; public surveys; generated regressors

JEL codes: C18, D14, G11, J31

*I thank Lorenz Kueng, Alberto Plazzi, Luise Eisfeld, Fabrizio Colella, Darrell Duffie, Moritz Kuhn, Petra Todd, Harald Uhlig, Sylvain Catherine, Jessica Wachter, and seminar participants for helpful comments. Earlier versions circulated as *Asymmetric Labor Income Risk: Implications for Risk-Taking in Financial Markets and Perceived Asymmetric Labor Income Risk and Household Portfolios*. The present version supersedes those drafts. All errors are my own.

[†]Università della Svizzera italiana, Via Giuseppe Buffi 13, Lugano, Switzerland. Email: tai.lo.yeung@usi.ch.

1 Introduction

Labor income is the largest nontradable asset of many households, yet “more labor-income risk” is incomplete when favorable and adverse tails move differently. A wider lower tail expands the set of earnings disasters; a wider upper tail expands the set of earnings opportunities. Variance and the 90–10 span register both changes as more risk even though households may respond to them in opposite directions. This paper asks whether separating the two tails changes what we learn about household portfolio choice (Viceira, 2001; Cocco, Gomes, and Maenhout, 2005; Chang, Hong, and Karabarbounis, 2018; Catherine, 2022; Catherine, Sodini, and Zhang, 2024; Shen, 2024; Gálvez and Paz-Pardo, 2026).

The main result is that lower and upper earnings tails organize portfolio entry in opposite directions within the baseline public-cell design, but the two sides are not equally stable. Among households that report no stock-and-mutual-fund holdings in the preceding year, a one-standard-deviation wider lower tail predicts a 2.71-percentage-point lower probability of positive holdings one year later; a wider upper tail predicts a 1.92-point higher probability. The pair remains jointly distinguishable from zero when 200 CPS resamples rebuild the first stage ($p = 0.0022$). A separate finite-cell check uses 99,999 null-imposed wild-cluster draws at the 60 CPS signal cells: the lower and upper coefficients have two-component adjusted q -values of 0.019 and 0.087, and the signed composition contrast has $p = 0.033$. An exact rotation leaves total 90–10 span imprecisely estimated and the lower-minus-upper composition contrast negative. Recomputing the early-CPS tails after excluding destination records classified as self-employed sharply attenuates the lower estimate. Across alternative zero-endpoint constructions, signed composition remains negative, while the standalone upper coefficient stays positive but is not always precise. The upper-tail sign also changes under an occupation-based assignment schema. The safe empirical content is therefore tail composition within the baseline public assignment, not two universally robust standalone tail effects.

I establish this result in four steps. First, I decompose public earnings histories into lower-tail distance, upper-tail distance, their total span, and a scale-free orientation measure. Second, I estimate these objects in CPS histories ending before the SIPP portfolio sample begins and ask whether they predict subsequent transitions into positive reported holdings. Third, I determine what the assigned tail measures capture by separating reproducibility from zero incidence, location, dispersion, and shape-specific variation. Finally, I rebuild the nonlinear CPS first stage and use individual subjective distributions as a separate construct check. This sequence puts the household-finance fact first and uses the measurement analysis to determine how strongly it can be interpreted.

The empirical design combines matched CPS Annual Social and Economic Supplement records with the Survey of Income and Program Participation. Baseline-positive earners are followed for one year, destination zeros are retained, and earnings growth is measured by the bounded arc rate. The CPS construction contains 183,809 transitions in public industry, education, and age cells. For the primary portfolio test, cell distributions use CPS origin years through 2017 and are assigned to SIPP portfolios observed from 2019 through 2024. The samples contain different people and

every portfolio outcome occurs after the final CPS destination year. This removes own-observation mechanical correlation and imposes strict temporal ordering, while leaving selection into earnings-risk cells intact.

The baseline entry pattern appears in three representations of the same tail information. Entering lower and upper distances jointly yields the opposite-sign coefficients above. Rotating them into total span and lower-minus-upper composition leaves span imprecise and composition negative. A scale-free incremental Kelley measure, constructed after separating zero incidence, mean growth, and dispersion, predicts a 1.05-percentage-point lower transition relative to an 8.79% entry rate and remains distinguishable from zero under the CPS plug-in sensitivity. The zero-endpoint exercises are more discriminating: the signed composition contrast remains negative throughout, the standalone upper coefficient remains positive but is imprecise once, and incremental Kelley is imprecise in two constructions (Section 4.3). These checks do not propagate CPS first-stage uncertainty. The corresponding association with the standing all-worker portfolio share also loses conventional significance once first-stage variation is added. The evidence is consequently strongest for tail composition and the transition to positive reported holdings within the baseline assignment.

That distinction has a direct household-finance interpretation. Fixed and per-period participation costs make risky-asset acquisition and retention state dependent even when conditional risky shares change little (Vissing-Jorgensen, 2002; Gomes and Michaelides, 2005; Fagereng, Gottlieb, and Guiso, 2017). Consistent with this logic, Giglio et al. (2021) finds that belief changes predict the direction and magnitude of trades conditional on trading more clearly than they predict when investors adjust. The outcome here is a transition from zero to positive observed stock-and-mutual-fund holdings, not a verified purchase or first-ever account opening. Appendix A.3 gives a deliberately limited benchmark: mirroring a centered earnings shock preserves its absolute-shock distribution and every even moment while reversing its orientation, and can change the optimal risky position in a one-period problem. This is a possibility result, not a mapping from the empirical quantile gaps into third moments, a calibration of entry, or evidence about the mechanism behind the estimated coefficients.

The measurement analysis explains why the corrected result is more informative than a raw skewness regression. Retaining transitions to zero turns a weakly reproducible positive-earner ranking into a reproducible unconditional downside ranking. In CPS, raw Kelley reliability is 0.740, but reliability falls to 0.370 after zero incidence, mean growth, and dispersion are separated; in the matched one-transition-per-person SIPP design, the corresponding values are 0.602 and 0.151. The raw ranking is therefore informative about a joint downside environment, not automatically a shape-specific primitive. The anatomy of the support boundary reinforces this point: CPS annual wage zeros often coincide with no work, whereas SIPP monthly main-job zeros mostly occur inside a contemporaneous work arrangement. Public data can rank survey-specific earnings environments, but the economically relevant orientation contrast must be stated conditionally.

Because the tail measures are estimated rather than observed, conventional clustering is not the end of inference. The primary resampling exercise redraws CPS people within signal cells and rebuilds quantiles, support, zero incidence, moments, residualization, standardization, assignment,

and the SIPP transition sample. Under the resulting plug-in sensitivity, the lower and upper distances remain jointly distinguishable from zero, and the scale-free entry coefficient has $p = 0.0052$ and a three-outcome $q = 0.0157$. Complementary wild-cluster inference draws bootstrap weights where the regressors vary—at the 60 CPS cells—while retaining two-way person and cell covariance (Roodman et al., 2019); the lower tail, upper tail at the 10% threshold, and signed composition each survive. The analogous standing-share result is imprecise after first-stage variation is added. The CPS and wild-cluster exercises address different uncertainties and are not a centered joint CPS–SIPP bootstrap, so they remain sensitivities rather than definitive joint inference. The strongest supported result is the baseline joint lower/upper test and its signed-composition rotation, not a standalone upper-tail coefficient that is invariant to every construction or grouping.

The paper contributes first to household finance. Existing work often summarizes labor-income risk with variance or emphasizes adverse exposure. Recent research shows that non-Gaussian earnings risk can matter for portfolios and life-cycle behavior (Catherine, Sodini, and Zhang, 2024; Shen, 2024; Gálvez and Paz-Pardo, 2026); close concurrent work also links perceived right-tail earnings opportunities to actual portfolios (van den Akker et al., 2026). The result here is different: lower- and upper-tail movements estimated from public histories enter jointly, predict a later portfolio transition with opposite signs, and imply an imprecise omnibus span. The exact span–composition rotation shows empirically how a one-number scale measure can conceal the contrast between opportunities and disasters. An exploratory common-sample comparison with five other asset-entry margins is not clean enough to establish outcome specificity: the bond vector differs in the all-worker frame, but that comparison weakens after selecting one worker per household and neither scalar contrast survives the full family adjustment. I therefore do not use the cross-asset exercise as a contribution or a mechanism test.

The second contribution concerns assigned measures of background risk. Empirical work either elicits households’ expectations (Guiso, Jappelli, and Terlizzese, 1996; Wang, 2023; Caplin et al., 2026; Arellano et al., 2026) or estimates risk from realized histories and assigns it across workers or samples (Angerer and Lam, 2009; Betermier et al., 2012; Chang et al., 2022; Fagereng, Guiso, and Pistaferri, 2018; Ranish, 2013; d’Astous and Shore, 2024). The public-history design scales broadly, but it makes construction and generated-regressor uncertainty part of the empirical object. I combine person-level reproducibility, incremental content, cross-survey transport, and nonlinear first-stage rebuilding for the same statistic. This extends the two-sample household-finance logic in Ranish (2013) and the broader lesson that construction choices can change downstream finance conclusions (Akey, Robertson, and Simutin, 2026).

The interpretation is descriptive. Historical cell distributions are assigned measures rather than elicited beliefs; annual growth is not decomposed into permanent and transitory shocks; group membership and the lagged-nonholder risk set are selected; and SIPP observes combined stock-and-mutual-fund balances rather than total equity. The transition family emerged during exploratory rather than preregistered analysis, so the reported false-discovery-rate families do not cover every specification considered during the project’s history. These limits qualify behavioral and causal claims.

Three further checks in Section 4.3 bound the pattern itself: within-person estimates are sensitive to whether tails are assigned from the lagged or current cell, the single pre-pandemic outcome year is too imprecise to corroborate or refute the primary estimates, and the upper-tail sign is not invariant to the grouping used to assign risk.

Section 2 defines the earnings-tail measures and empirical design. Section 3 describes the public samples and strict out-of-time assignment. Section 4 presents the household-finance evidence. Section 5 then shows what the assigned tails capture. Section 6 rebuilds the CPS first stage and reports additional construct checks. Section 7 draws implications for measuring labor-income risk, and Section 8 concludes. Detailed robustness results and the SCE timing analysis are in the Online Appendix.

2 Earnings-Tail Measures and Empirical Design

2.1 Disasters, opportunities, and orientation

Let G_c denote the distribution of one-year earnings growth for workers in cell c , and let $q_{p,c}$ be its p th quantile. I orient every asymmetry measure so that a positive value denotes a longer or heavier downside. The upper and lower 90–50 gaps are

$$U_c = q_{0.90,c} - q_{0.50,c}, \quad L_c = q_{0.50,c} - q_{0.10,c}. \quad (1)$$

Their sum $S_c = L_c + U_c$ is the 90–10 spread and their difference $D_c = L_c - U_c$ is an unscaled tail imbalance. The primary scale-free statistic is downside-oriented Kelley shape,

$$K_c = \frac{L_c - U_c}{L_c + U_c} = \frac{2q_{0.50,c} - q_{0.10,c} - q_{0.90,c}}{q_{0.90,c} - q_{0.10,c}}. \quad (2)$$

Bowley shape replaces the 10th and 90th percentiles with the 25th and 75th percentiles. I also consider downside-oriented L-skewness, conventional standardized third-moment skewness with its sign reversed, and probability imbalance beyond one within-cell standard deviation.

These estimators answer different questions. Tail gaps measure magnitudes; tail-probability balance measures mass; Kelley and Bowley normalize by an interquantile range; moment skewness is sensitive to the full support. The paper distinguishes three economic objects. The *joint downside environment* is the unconditional distribution G_c , including transitions to zero measured earnings. *Conditional wage-growth shape* is the distribution among workers with positive earnings at both endpoints. *Incremental tail orientation* is the component of a stated tail statistic that varies within the observed support after the zero-transition probability, mean growth, and dispersion are partialled out. The first two objects are distributions. The third is a conditional empirical contrast, not a primitive structural parameter.

A destination-zero mass links the first object to lower moments exactly.

Lemma 1 (A point mass below the positive-earnings support). *Let F have support strictly above a , finite mean μ_F , and variance σ_F^2 . For $p \in [0, 1]$, let $G_p = p\delta_a + (1 - p)F$. Using generalized-inverse quantiles, for $\tau \in (0, 1)$,*

$$q_\tau(G_p) = \begin{cases} a, & \tau \leq p, \\ F^{-1}((\tau - p)/(1 - p)), & \tau > p, \end{cases}$$

while

$$\mu(p) = pa + (1 - p)\mu_F, \quad \text{Var}(G_p) = (1 - p)\sigma_F^2 + p(1 - p)(\mu_F - a)^2.$$

Consequently, cross-cell variation in a stable destination-zero share changes the quantile map, mean, and dispersion even when the positive-earnings distribution F is identical. The resulting Kelley statistic remains a legitimate description of G_p . What it cannot do alone is identify a counterfactual comparison that changes tail orientation while holding destination-zero incidence and lower moments fixed. Lemma 1 does not imply that Kelley shape is monotone in p without additional restrictions on F . Appendix A.1 gives the proof.

2.2 Earnings support and growth measurement

Log growth requires positive earnings in both periods. Applying it to continuously positive earners estimates the distribution of wage growth conditional on positive measured earnings at both endpoints. That object can be useful, but it excludes transitions to zero measured earnings, which can reflect job loss, nonemployment, changes in work intensity or work arrangement, and survey-specific earnings definitions. Labor-market transitions are economically relevant to perceived earnings risk (Caplin et al., 2026), but the anatomy of a recorded zero must be shown rather than inferred from the endpoint alone. Adding one dollar before taking logs retains zeros numerically but imposes a unit-dependent distance between zero and small positive earnings.

The baseline therefore uses bounded arc growth (Törnqvist, Vartia, and Vartia, 1985),

$$g_{i,t+1} = 2 \frac{y_{i,t+1} - y_{i,t}}{y_{i,t+1} + y_{i,t}}, \tag{3}$$

where baseline earnings are positive and destination earnings may be zero. The rate is symmetric in the two earnings levels, lies in $[-2, 2]$, and maps a destination zero to -2 without an arbitrary offset. Support choice is part of the estimand: positive-to-positive log and arc growth are reported as alternative objects, not as cleaning variations around the same object.

2.3 Measurement reliability

For each construction, I assign every unique person globally to one of two halves, estimate cell statistics separately, and calculate the correlation $r_{1/2}$ across cells. Global assignment prevents the same worker from entering both halves after an industry or education change. The full-sample

reliability projection is the Spearman–Brown transform (Spearman, 1910; Brown, 1910)

$$\rho_{\text{SB}} = \frac{2r_{1/2}}{1 + r_{1/2}}. \quad (4)$$

The reported value applies equation (4) to the mean half-sample correlation across random person splits. I also report rank correlation and early-versus-late stability. Support is measured using the Kish effective sample size (Kish, 1965), $N_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$, not the raw number of records.

Reliability has a narrow interpretation. The following decomposition makes the limit explicit.

Lemma 2 (Reliability of a composite statistic). *Suppose the two independent half-sample estimates of a group statistic satisfy*

$$\hat{K}_c^{(h)} = a'Z_c + s_c + e_c^{(h)}, \quad h \in \{1, 2\},$$

where Z_c is a stable vector of lower distributional features, s_c is stable incremental tail variation, and the half-sample errors are mutually independent and independent of (Z_c, s_c) . If their error variances are equal to σ_e^2 , then

$$\text{Corr}(\hat{K}_c^{(1)}, \hat{K}_c^{(2)}) = \frac{\text{Var}(a'Z_c + s_c)}{\text{Var}(a'Z_c + s_c) + \sigma_e^2}.$$

Reliability can therefore converge to one as sampling error vanishes even when $\text{Var}(s_c) = 0$.

The result does not declare $a'Z_c$ spurious. It shows that reliability cannot determine which stable economic feature produces the ranking. That determination requires a stated conditional contrast.

2.4 Incremental content

I estimate a sequential projection across supported cells. The most demanding specification is

$$K_c = a + \gamma p_{0c} + f_2(\mu_c) + h_2(\sigma_c) + \lambda_{e(c)} + \lambda_{a(c)} + u_c, \quad (5)$$

where p_{0c} is the transition-to-zero probability; f_2 and h_2 are quadratic functions; and the fixed components capture education and age bands. The fit is weighted by effective cell support capped at 5,000. I report the raw statistic, the residual after the linear p_{0c} term and demographic components, and the residual after the complete projection. Equation (5) is re-estimated within each person half. Residualizing the full-sample statistic once and then splitting would leak information across the reliability test.

Including mean growth is particularly important given the strong empirical association between location and Kelley shape documented by Harmenberg (2021). The projection defines a conditional contrast; it does not declare location or scale economically spurious.

The residual u_c is an operational incremental signal, not “true skewness.” Equation (5) does not recover a latent shock process, prove that correlated distributional features are confounds, or define a universally preferred shape estimator. It asks the narrower question required by a shape-specific portfolio interpretation: after allowing a transparent relation with the positive-measured-earnings

boundary, location, and scale, is enough stable cross-cell variation left to support a downstream regression?

2.5 Portfolio assignment and inference

Let $\hat{u}_c^{\text{CPS}}$ be the standardized incremental signal estimated outside the SIPP. For worker i in SIPP year t , the portfolio application estimates

$$P_{it} = \beta \hat{u}_{c(i)}^{\text{CPS}} + \rho \hat{p}_{0,c(i)} + f_2(\hat{\mu}_{c(i)}) + h_2(\hat{\sigma}_{c(i)}) + X'_{it}\gamma + \delta_t + \delta_j + \delta_e + \delta_a + \varepsilon_{it}. \quad (6)$$

Controls include log income, log wealth, age and age squared, sex, and an unemployment-spell indicator; fixed effects cover year, broad industry, education, and age band. SIPP person-year weights are used. Conditional standard errors are two-way clustered by SIPP person and CPS signal cell; the CPS-first-stage sensitivity also resamples the independent CPS measurement sample and rebuilds equation (5). Coefficients are descriptive associations per one cross-cell standard deviation of $\hat{u}_c^{\text{CPS}}$. Group assignment is endogenous, and equation (6) is not interpreted causally.

The dynamic application defines $H_{it} = 1\{A_{it} > 0\}$, where A_{it} is the observed stock-and-mutual-fund balance. For consecutive annual observations, signed participation change is $\Delta H_{it} = H_{it} - H_{i,t-1}$; entry is H_{it} in the risk set $H_{i,t-1} = 0$; and exit is $1 - H_{it}$ in the risk set $H_{i,t-1} = 1$. The signal and all cell-distribution controls are assigned from the lagged observation. Household controls and fixed effects are likewise lagged except for the current outcome-year effect, and the current SIPP weight is used. Both observations must lie in 2019–2024, so transition outcomes span 2020–2024. The entry outcome is a nonholder-to-holder transition in observed balances, not necessarily a household’s first-ever stock-market entry.

3 Public Data and Out-of-Sample Timing

3.1 Earnings-history samples

The CPS ASEC supplies annual wage and salary income, stable person identifiers across adjacent surveys, industry, education, age, and survey weights. The raw extract covers interview years 2014–2023. I begin with workers aged 25–65 who have positive baseline wage earnings and match them to the next ASEC observation, allowing destination earnings to equal zero. Recorded age must rise by zero, one, or two years. The exact CPS-to-SIPP crosswalk yields 183,809 unique-person transitions in 72 candidate broad-industry-by-education-by-age cells. Education code 73, a high-school diploma in the IPUMS CPS coding, is included in the high-school group. Age is split at 45. Appendix C records the industry and education mapping, support rules, and sample waterfall.

The post-redesign SIPP is a public longitudinal survey with retrospective monthly earnings and annual household balance-sheet measures. For the independent measurement comparison, I match a positive monthly main-job earnings observation to the same calendar month 12 months later and

allow the destination value to be zero. Destination-month job-or-business class and no-job-spell fields reveal whether a main-job zero occurs inside an employer, self-employment, or other arrangement and whether any no-job spell is recorded during that month. Repeated observations from one worker always remain in the same split half. Because overlapping monthly transitions may mechanically stabilize cell estimates, the primary SIPP reliability check also keeps one fixed nonoverlapping 12-month-apart transition per person. The repeated-month design is reported as a precision benchmark rather than treated as independent information.

Both surveys use their official person weights to estimate distributions. Effective support is measured by the Kish count. Reliability splits assign each person globally to one half; the CPS first-stage sensitivity instead resamples person clusters within signal cell and is described separately in Section 6. The baseline construction retains destination zeros and uses equation (3); positive-to-positive arc and log growth define separate conditional-wage objects. The broader estimator and support tournaments are in Appendix B.

3.2 Later portfolio levels and transitions

The primary finance design estimates the group distribution using CPS transitions whose origin year is no later than 2017 and assigns that frozen ranking to SIPP portfolios from 2019 through 2024. The crosswalk uses 12 harmonized nonmilitary broad industries, three education groups, and two age bands. Because the surveys contain different people, no SIPP portfolio record enters the measured CPS distribution. CPS destination earnings extend through 2018, so every primary portfolio outcome strictly postdates the measurement window. A broader-window robustness check restores SIPP portfolios from 2018. Neither design makes group membership exogenous or proves that households perceived the assigned history.

Monthly SIPP balance-sheet observations are collapsed to person-years by averaging the available eligible records within each calendar year. Participation equals one when the observed stock-and-mutual-fund balance is positive. The primary share sets the observed balance to zero for nonparticipants and divides it by reported financial assets. Conditional-positive-asset and holder shares are reported secondary outcomes. SIPP does not observe the allocation inside retirement accounts, mutual funds can contain non-equity securities, and employer-stock compensation cannot be cleanly removed. The outcome is therefore labeled the *observed stock-and-mutual-fund asset share*, not total equity exposure. Appendix C.3 reports denominator support, ratio bounds, eligible-month coverage, and repeated-household-record diagnostics for the exact estimation frame.

The transition frame links consecutive annual observations and assigns the lagged early-CPS signal to the next year's balance state. It contains 35,112 person-year pairs on 23,170 people: 27,752 observations begin with zero holdings and 7,360 begin with positive holdings. The weighted nonholder-to-holder transition rate is 8.79%. Requiring the same signal cell in both years is a migration sensitivity, not part of the primary risk set. Sustained entry additionally requires positive holdings one year after the transition; nominal \$1,000 and \$5,000 crossings test whether a small positive balance alone drives the result.

Every regression uses SIPP person-year weights and household controls measured at the relevant date. Conditional inference clusters by person and assigned CPS signal cell; a cell-year aggregation and a CPS first-stage resampling sensitivity address the limited number of distinct signals. Appendix F separately applies the measurement logic to public subjective earnings distributions from the New York Fed Survey of Consumer Expectations.

4 Asymmetric Earnings Tails and Portfolio Entry

4.1 Main estimates

Design. The primary application uses 60 early-CPS signal cells assigned to 90,110 SIPP worker-years on 52,115 people observed from 2019 through 2024. The headline risk set contains 27,752 consecutive-year transitions that begin with no reported stock-and-mutual-fund holdings. Every specification conditions on the cell zero share, mean growth, dispersion, household covariates, and fixed effects. Standard errors cluster by SIPP person and assigned CPS cell; the displayed plug-in inference also rebuilds the CPS first stage.

Lower and upper tails. Table 1 separates the entry coefficient into its two economic sides. The lower distance is $L_c = q_{0.50,c} - q_{0.10,c}$ and the upper distance is $U_c = q_{0.90,c} - q_{0.50,c}$. Panel A enters their standardized early-CPS values jointly. Panel B rotates the same raw column space into total span $S_c = L_c + U_c$ and signed composition $D_c = L_c - U_c$; fitted values agree to 1.1×10^{-8} . Panel C conditions the scale-free Kelley orientation measure directly on total span. All panels use the same risk set and empirical design.

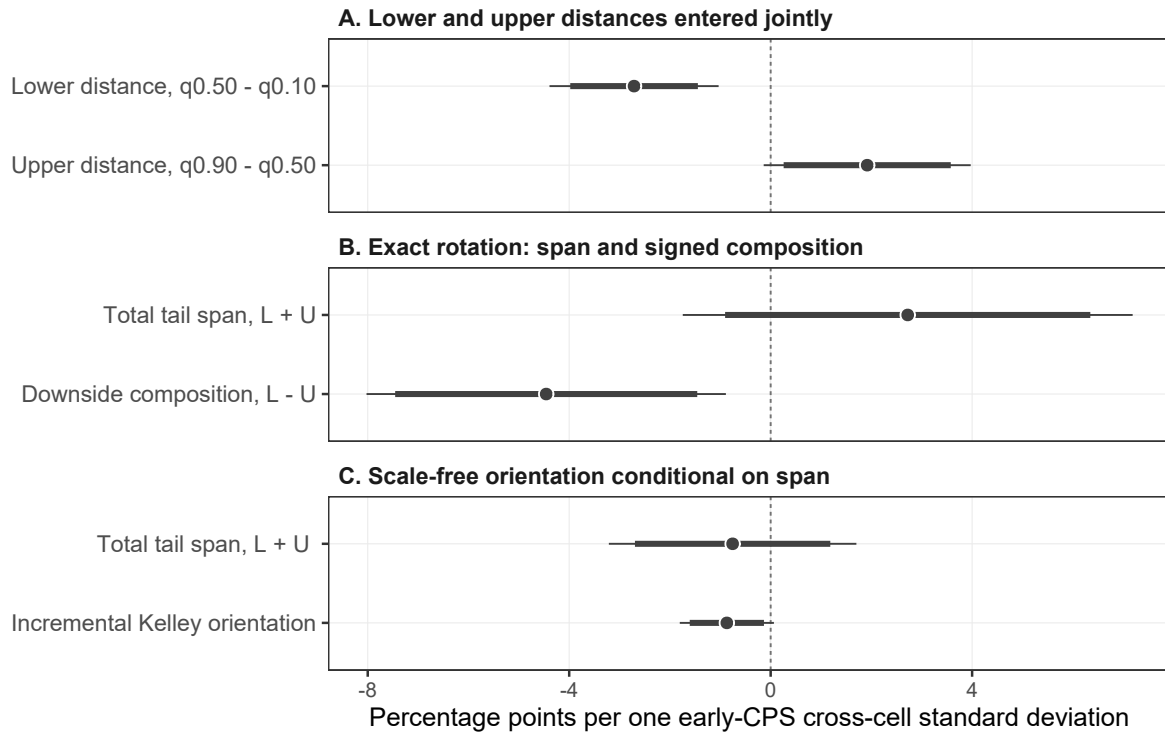
Portfolio entry. Among lagged nonholders, incremental Kelley predicts a 1.047-percentage-point lower probability of positive observed holdings one year later (SE 0.272 point; $p = 0.00030$; three-outcome $q = 0.00089$). Relative to the weighted 8.79% entry rate, the point estimate is 11.9%. Raw Kelley gives a similar point estimate but less precision: its unadjusted $p = 0.042$ has a three-outcome $q = 0.126$. The tail decomposition supplies the economic content that a one-index regression cannot. Conditional on each other, lower- and upper-tail distances have opposite signs. In the plug-in calculation that rebuilds the CPS first stage, their joint p -value is 0.0022; lower-tail distance has $p = 0.0015$, while the upper-tail estimate is suggestive on its own ($p = 0.068$). In the 99,999-draw cell-level wild bootstrap, the corresponding lower and upper BH q -values are 0.019 and 0.087, and signed composition has $p = 0.033$. In the CPS plug-in calculation, signed composition has $p = 0.014$, whereas total span is imprecise. Figure 1 makes this contrast visible. Section 4.3 shows that the lower-tail precision depends on the zero-endpoint construction. Across those conditional checks, signed composition remains negative and distinguishable from zero, while the upper estimate remains positive but is imprecise once; an alternative occupation-based grouping reverses its sign. The supported result is therefore the baseline joint test and signed-composition rotation, not an upper-tail coefficient invariant to every construction or assignment schema.

Scale versus composition. Assigned cell dispersion is also imprecisely estimated once tail composition

Table 1. Lower and upper earnings tails and portfolio entry

Early-CPS distributional component	Coef.	Cond. S.E.	CPS S.D.	Plug-in S.E.	<i>p</i>
<i>Panel A. Lower and upper tails</i>					
Lower distance, $q_{0.50} - q_{0.10}$	-2.712***	0.646	0.562	0.857	0.002
Upper distance, $q_{0.90} - q_{0.50}$	1.917*	0.846	0.619	1.049	0.068
<i>Panel B. Total span and tail composition</i>					
Total tail span, $L + U$	2.721	1.850	1.331	2.279	0.232
Downside composition, $L - U$	-4.456**	1.530	0.986	1.820	0.014
<i>Panel C. Total span and scale-free orientation</i>					
Total tail span, $L + U$	-0.755	0.989	0.770	1.254	0.547
Incremental downside Kelley orientation	-0.870*	0.376	0.293	0.477	0.068

Notes: The outcome is a one-year transition from zero to positive reported stock-and-mutual-fund holdings. Each panel jointly includes the two displayed standardized CPS measures and the lagged controls and fixed effects described in the text. Coefficients are percentage points per one cross-cell standard deviation; conditional standard errors cluster by SIPP person and CPS signal cell. "CPS S.D." is the coefficient standard deviation across 200 CPS resamples, and plug-in standard errors add this component to the conditional variance. Stars use plug-in two-sided *p*-values: ****p* < 0.01, ***p* < 0.05, and **p* < 0.10. In a separate 99,999-draw null-imposed wild-cluster check, lower and upper distances have two-component BH *q*-values of 0.019 and 0.087; signed composition has *p* = 0.033. The sample contains 27,752 transitions, 19,300 people, and 60 signal cells. Assignment is endogenous.

Figure 1. Tail decomposition of portfolio entry

Notes: Points are the coefficients of Table 1, in percentage points per one early-CPS cross-cell standard deviation. Thick intervals use conditional two-way-clustered standard errors; thin intervals add the 200-draw CPS first-stage component. The dashed line marks zero. Components are standardized separately within panel and enter jointly.

Table 2. Assigned tail orientation, portfolio levels, and transitions

Outcome	Raw Kelley			Incremental Kelley			Obs.
	Coef.	S.E.	Cond. q	Coef.	S.E.	Cond. q	
<i>Panel A. Standing outcomes in 2019–2024</i>							
Participation	0.38	0.73	0.880	−0.92*	0.48	0.081	90,110
Share, positive financial assets	0.061	0.185	0.880	−0.261***	0.096	0.035	84,161
Share, all workers	0.080	0.185	0.880	−0.234**	0.097	0.039	90,110
Share, current holders	−0.097	0.638	0.880	−0.153	0.424	0.721	17,578
<i>Panel B. Consecutive-year transitions; both observations in 2019–2024</i>							
Signed participation change	−0.335	0.428	0.437	−0.731***	0.260	0.010	35,112
Nonholder to positive holdings	−1.093**	0.526	0.126	−1.047***	0.272	0.001	27,752
Holder to zero holdings	−2.236	1.538	0.227	−0.813	1.152	0.483	7,360

Notes: Coefficients and standard errors are percentage points per one cross-cell standard deviation. Incremental specifications include the CPS zero share, quadratic mean and dispersion, household controls, and year, industry, education, and age-band fixed effects. Panel B assigns lagged signals and controls and uses lagged-state risk sets. Standard errors are two-way clustered by person and CPS signal cell. Stars use unadjusted conditional two-sided tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. Benjamini–Hochberg q -values adjust the four standing outcomes or three transitions within specification (Benjamini and Hochberg, 1995). All outcomes strictly postdate CPS destination earnings through 2018.

is made explicit. In the locked entry frame, the standardized linear term in cell standard deviation, evaluated at mean standardized dispersion, is −0.262 point (SE 1.120; $p = 0.816$) when entered with its square and the same controls. Adding Kelley orientation or separate lower and upper distances does not reveal a positive scale coefficient. The current evidence therefore supports opposing conditional associations for the two tails, not a sign reversal in a variance coefficient. Signed participation change is negative for incremental Kelley; exit remains too imprecise to characterize.

Standing outcomes. Table 2 contrasts the transition with accumulated holdings. Incremental Kelley is negative for all four standing outcomes, but the all-worker share’s conditional $p = 0.019$ and four-outcome $q = 0.039$ become plug-in $p = 0.109$ after the CPS first stage is rebuilt. The broader 2018–2024 window is in Appendix Table E.1; household-level clustering and alternative decision-unit restrictions preserve the negative conditional direction (Appendix Table C.3). The supported result is consequently strongest on entry rather than on portfolio levels generally.

Robustness. Requiring the same signal cell in both years gives −0.958 point ($p = 0.0039$; three-outcome $q = 0.0118$). Omitting each outcome year in turn yields five negative coefficients from −1.203 to −0.881 point. Sustained entry—zero holdings at $t - 1$ and positive holdings at both t and $t + 1$ —is −0.697 point (SE 0.293; six-test $q = 0.041$). Nominal crossings above \$1,000 and \$5,000 are −1.104 and −0.453 point, with six-test $q < 0.001$ and $q = 0.045$. The stable-household frame gives −1.057 point, and selecting one highest-earning eligible worker per household transition gives −0.869 point. Every leave-one-cell and leave-one-industry coefficient is negative; cell-year aggregation and nonlinear zero-incidence controls produce similar estimates. A formal common-sample comparison with bank, bond, retirement, home-equity, and vehicle-equity entry distinguishes only the bond coefficient vector in the all-worker frame ($q = 0.017$). Selecting one highest-earning worker per lag household weakens its adjusted joint evidence to $q = 0.102$, and neither lower nor upper scalar

contrast survives the ten-coefficient family. The exercise is therefore an exploratory cross-outcome diagnostic, not evidence of outcome specificity, a placebo, or a mechanism. Appendix Tables C.4, C.6, and C.7 report the complete displayed families.

Interpretation. The transition family and tail decomposition emerged from exploratory rather than preregistered analysis; the reported q -values cover the displayed outcome families, not every specification considered during the project’s history. Risk-tolerant workers can sort into both volatile jobs and risky portfolios (Ranish, 2013; d’Astous and Shore, 2024), and historical group distributions need not equal household beliefs (Guiso, Jappelli, and Terlizzese, 1996; Wang, 2023; Caplin et al., 2026; Arellano et al., 2026). The estimates therefore describe a transition to positive observed holdings rather than a first-ever account opening or a behavioral response to an exogenous change in perceived risk.

4.2 Interpretation and precision

Incremental Kelley reliability is 0.397 for the exact 60-cell early signal assigned to portfolios, 0.370 in pooled CPS, and 0.151 in matched SIPP. These values make attenuation a first-order concern. A coefficient on the measured signal is not an estimate of the effect of latent tail orientation corrected for measurement error, and its confidence interval is not an equivalence bound for such an effect. The effective identifying variation is 60 early-CPS cells, not tens of thousands of independent risk environments. The cell-level wild bootstrap addresses finite-cluster inference directly and preserves the tail-composition result; it does not increase the number of independent environments. A normal-approximation power calculation shows that the standing level family is underpowered once first-stage variation is included: at 5% size and 80% power, the measured-signal minimum detectable effects are 0.409 point for the all-worker share and 1.79 points for participation under the CPS plug-in. Appendix Table G.1 reports the full calculation. Entry remains distinguishable from zero under that plug-in sensitivity, but neither inference exercise repairs endogenous assignment, corrects attenuation, or converts the measured signal into latent risk.

Pooled estimates, the broader construction tournament, alternative centered estimators, and the SCE construct and timing checks are in the Online Appendix. They do not supply causal variation or override the hierarchy in Table 2.

4.3 Where the tail coefficients come from

The lower-tail distance is not a smooth object across cells. In 11 of the 60 signal cells the destination-zero share exceeds ten percent, so the 10th percentile sits at the arc-growth bound of -2 and L_c equals $2 + q_{0.50,c}$; across cells the standardized lower distance correlates 0.89 with the zero share and 0.77 with an indicator for the bound. Table 3 therefore re-estimates the entry decomposition under three alternative constructions of the early-CPS tails. Recomputing those tails after excluding destination records classified as self-employed (6.2 percent of early transitions), or counting a destination zero only when no weeks were worked (which removes the 2.3 percent of transitions

that are worked-without-wage zeros), cuts the cross-cell dispersion of the lower distance from 0.40 to 0.25 and 0.23 and reduces the lower coefficient from -2.71 to -0.91 and -0.76 points, neither distinguishable from zero. Restricting the sample to the 49 cells whose 10th percentile is interior gives -1.22 points (SE 1.35).

Table 3. Where the lower-tail coefficient comes from: alternative zero-endpoint constructions

Construction of the early-CPS tails	SD of L	Lower L	Upper U	$L - U$	Incr. Kelley	Cells
A. Baseline: all linked transitions	0.398	-2.712^{***} (0.646)	1.917^{**} (0.846)	-4.456^{***} (1.530)	-1.047^{***} (0.272)	60
B. Exclude self-employed destination records	0.249	-0.907 (0.887)	1.392 (0.902)	-1.730^{**} (0.857)	-0.325 (0.230)	60
C. Zero only if no weeks worked	0.231	-0.758 (0.895)	1.660^{**} (0.817)	-1.767^{**} (0.757)	-0.432^* (0.243)	60
D. Baseline, interior-percentile cells only	—	-1.215 (1.352)	2.306^{**} (1.114)	-4.458^{**} (1.845)	-0.683^{**} (0.337)	49

Notes: The outcome is the one-year transition from zero to positive reported stock-and-mutual-fund holdings in the locked 27,752-observation risk set (23,858 observations in row D). Coefficients are percentage points per one cross-cell standard deviation of the named early-CPS component, with conditional two-way-clustered standard errors by SIPP person and lagged CPS cell in parentheses. Each row rebuilds the early-cell quantiles, zero share, mean, and dispersion under its construction, re-standardizes across the retained cells, and enters the same lagged controls and fixed effects as Table 1. L and U are entered jointly; $L - U$ is entered with the total span; incremental Kelley is the residual of equation (5) under the same construction. “SD of L ” is the Kish-weighted cross-cell standard deviation of the lower distance in arc-growth units. Row B excludes transitions whose destination class of worker is self-employed (CPS CLASSWKR 13 or 14); row C removes destination wage zeros with positive weeks worked; row D keeps the baseline construction but drops the 11 cells whose 10th percentile equals the support bound. Row A reproduces the frozen cell measures and Table 1 exactly. Stars use unadjusted conditional two-sided tests: $***p < 0.01$, $**p < 0.05$, and $*p < 0.10$. These checks do not propagate CPS first-stage uncertainty.

The signed composition contrast is the most stable object across these zero-endpoint exercises: it is -1.73 , -1.77 , and -4.46 points in B–D (conditional $p = 0.048$, 0.023 , and 0.020). The upper estimate remains positive at 1.39 , 1.66 , and 2.31 points but is imprecise in B ($p = 0.13$); incremental Kelley remains negative at -0.33 , -0.43 , and -0.68 point but is imprecise in B and C ($p = 0.16$ and 0.080). Recomputing the early-CPS tails after excluding destination records classified as self-employed therefore attenuates the lower estimate and qualifies its interpretation as disaster risk. None of these construction checks propagates CPS first-stage uncertainty.

An exploratory business-asset diagnostic distinguishes a self-employment classification in the CPS first stage from business ownership in the SIPP portfolio. Among worker transitions whose household reports neither stock-and-mutual-fund holdings nor business equity at the lag date, a wider lower tail predicts 2.99 percentage points less stock-and-mutual-fund-only entry (CPS plug-in $p = 0.0012$; eight-coefficient $q = 0.0098$); the upper-tail estimate is $+2.28$ points but has $q = 0.131$. Neither business-only entry nor a broader business-entry outcome survives the same adjustment, and all corrected results disappear when the tails exclude self-employed CPS destinations or worked-without-wage zeros. The exercise therefore reinforces the baseline lower-tail entry margin but does not support entrepreneurial substitution or crowd-out as its mechanism.

Three further checks bound the interpretation (Appendix D). First, the within-person exercises are

unstable rather than uniformly null. Adding changes in assigned tails leaves the change coefficients near zero, and current-cell assignment attenuates the full-frame estimates. Among 5,789 workers with repeated entry-risk transitions, person fixed effects give -0.58 (SE 1.30) for the lower tail and 4.27 (SE 1.99) for the upper tail under lagged assignment, but -0.05 (SE 0.96) and -1.17 (SE 0.75) under current assignment. The reversal across assignment timing does not support a coherent within-person interpretation, and time-invariant sorting remains possible. Second, the 6,033 transitions from 2018 to 2019 give -0.34 and -0.46 points (SE 1.08 and 0.99). This single pre-pandemic holdout is too imprecise to corroborate or refute the locked estimates; pooling it with the locked frame gives -2.42 and 1.57 points. Third, split-sample heterogeneity by lagged liquid assets places the lower coefficient in the top tercile (-6.20 , SE 1.68) and the upper coefficient in the bottom tercile (1.98 , SE 0.88), a pattern that the descriptive design does not sharply interpret; interaction tests with common controls do not reject equal coefficients ($p = 0.38$ for the lower and 0.89 for the upper tail).

Changing the assignment schema further limits generality. Adding occupation to the industry-based schema shrinks the upper estimate to 0.32 point in 69 cells and 0.55 point in 96 cells as incremental reliability falls to 0.32 and 0.18 . Replacing industry with occupation in an 18-cell occupation-by-education-by-age schema reverses the upper estimate to -1.74 points (SE 0.98). The upper-tail sign is therefore not invariant to the grouping used to assign risk. Separately, this exercise does not validate the assigned history as a measure of perceived risk: in 143 CPS state-by-education-by-age cells matched to 75,863 SCE responses, a wider assigned upper tail predicts a 0.07 -standard-deviation narrower, rather than wider, perceived upper distance, while the assigned lower tail has no detectable counterpart (Appendix Table D.3).

Taken together, the checks make signed composition the most stable object within the baseline cell schema and zero-endpoint exercises. They do not establish a universally robust standalone upper-tail or Kelley coefficient, and they identify the support boundary as a feature that must be disclosed before the lower distance is read as disaster risk. I next ask which features of public earnings histories make the assigned tail measures reproducible and interpretable.

5 What the Assigned Earnings Tails Capture

5.1 The role of the support boundary

Figure 2 begins with the population choice. In a companion CPS construction comparison with the same age-refined cell schema, raw Kelley reliability is 0.018 when arc growth is restricted to positive earnings at both endpoints and 0.735 when destination zeros are retained. The exact crosswalk used for portfolio assignment gives 0.740 . In SIPP, keeping one 12-month-apart transition per person gives reliability 0.147 for positive-to-positive arc growth and 0.602 when destination zeros are retained. Positive-to-positive log growth is similarly weak at 0.120 . Repeated overlapping SIPP months are not responsible for the ordering: the zero-inclusive estimate is 0.622 with repeated months and 0.602

with one transition per person.

The construction reversal is economically interpretable. A transition to zero maps to the arc-growth lower bound and changes the unconditional earnings distribution. Conditioning it away estimates wage growth among workers with positive measured earnings at both endpoints, not the same risk object with fewer observations. The positive result is therefore that, within each survey, independent worker splits reproduce a downside-support ranking that includes transitions to zero measured earnings.

5.2 What a recorded zero contains

Table 4 opens the endpoint rather than assigning it an economic label. CPS wage-and-salary income and weeks worked refer to the same destination calendar year. Of 11,828 annual wage-zero destinations, 64.50% report no work, but 35.50% report some work and 26.02% report all 52 weeks. The two components are separately stable across independent people: Spearman–Brown reliability is 0.840 for no-work incidence and 0.940 for worked-without-wage incidence, compared with 0.940 for total zero incidence. The full-year-work-without-wage component has reliability 0.914. Thus the raw tail ranking is not just a measure of full-year nonwork; worked-without-wage incidence is at least as reproducible as total wage-zero incidence.

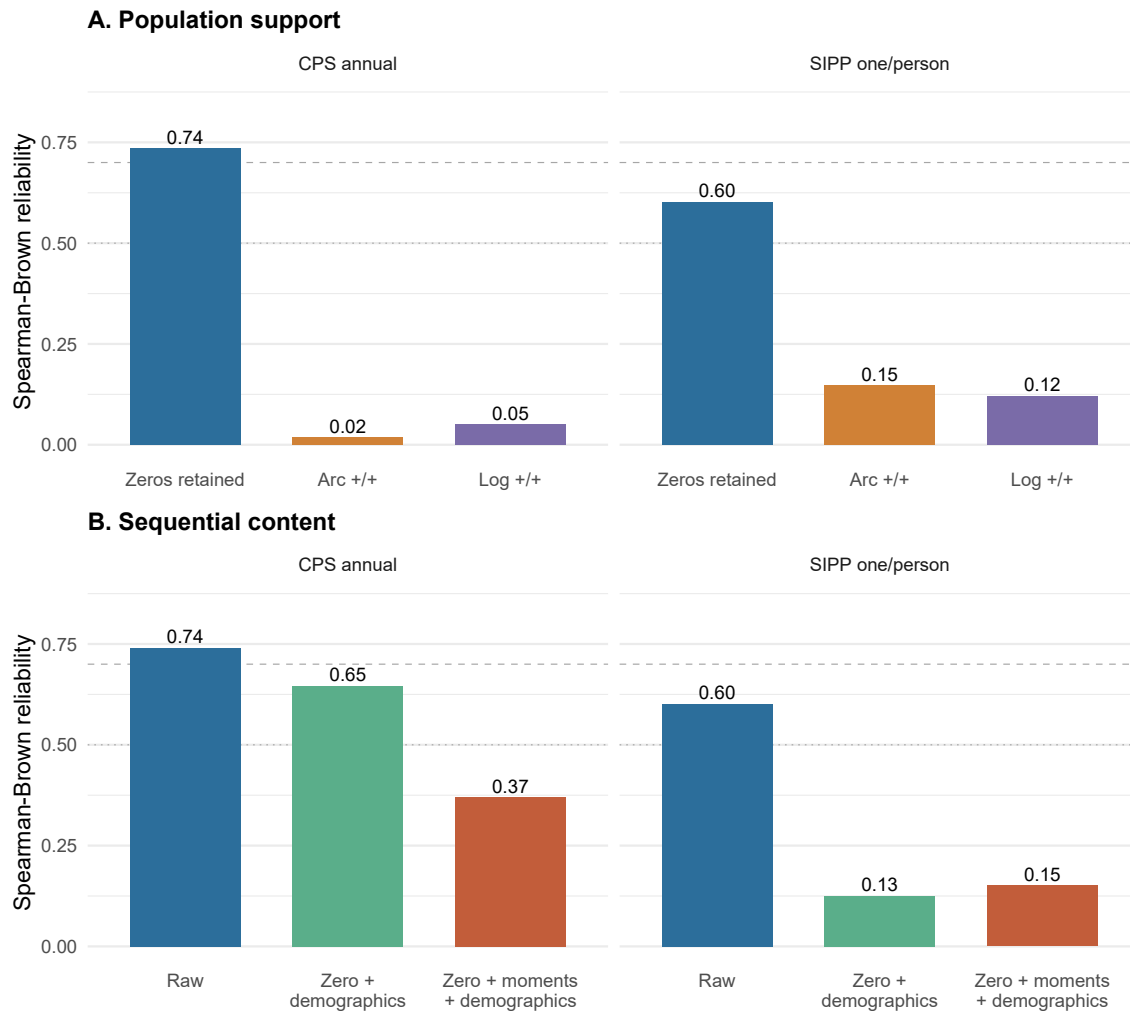
The SIPP endpoint is different again. In a strict out-of-time frame, one eligible 2018–2023 origin transition is selected uniformly per person without conditioning on destination earnings or job state; 914 of 25,310 selected workers enter zero monthly main-job earnings. Among those zeros, 64.47% have a contemporaneous self-employment arrangement and 34.62% an employer arrangement. For the former, the job-1 field can record zero business earnings rather than wage loss. Only 19 zeros, or a weighted 2.06%, contain any recorded no-job spell; the corresponding weighted share is 0.99% when all eligible overlapping transitions are retained. Arrangement class is mutually exclusive, while the within-month no-job flag can coexist with an arrangement. A numerical zero therefore carries economic content, but not one invariant event across the two public surveys.

An out-of-time prediction exercise is directionally consistent but does not survive the displayed multiplicity adjustment. Across 30 deterministic choices of one 2018–2023 origin transition per person, the early-CPS raw Kelley coefficient for later SIPP main-job-zero incidence is positive in 29, with median 0.297 percentage point and P10–P90 range 0.068–0.450, but no selection has a four-outcome-adjusted $q < 0.05$. The incremental coefficient is positive in 26 selections, with median 0.154 point and P10–P90 range -0.013 – 0.338 ; again none passes the adjusted threshold. Appendix Table B.5 reports the complete selection comparison. I therefore use the exercise as directional evidence about what the raw composite travels with, not as a new predictive rejection.

5.3 Incremental content and time stability

Table 5 asks what variation supports that ranking. The zero-transition share is highly reproducible: 0.940 in pooled CPS and 0.809 in the matched SIPP design. In pooled CPS, adding the zero share

Figure 2. Where the reproducible downside ranking comes from



Notes: Panel A compares raw downside-oriented Kelley reliability when destination zeros are retained with positive-to-positive constructions. Panel B uses the pooled exact CPS crosswalk and matched SIPP: the second stage removes the linear zero share and demographic components, and the third also removes quadratic mean and dispersion. Reliability is the Spearman–Brown transform of the mean independent-person half-sample correlation across 30 deterministic splits. Each projection is re-estimated inside every half.

and demographic components changes reliability from 0.740 to 0.646; adding quadratic mean and dispersion terms lowers it to 0.370. The exact early signal later assigned to portfolios has reliability 0.687 raw and 0.397 after the complete projection. In matched SIPP, the corresponding sequence is 0.602, 0.125, and 0.151. Thus the survey-specific zero endpoint and lower moments account for reproducible CPS variation, while the fully incremental SIPP ranking remains weak under either adjustment.

Within-survey time stability is a separate gate. The CPS early-versus-late correlation is 0.631 for raw Kelley and 0.230 for the full incremental signal. In the matched SIPP design it falls from 0.414 to -0.100 . Rank correlations are 0.634 and 0.326 in CPS, and 0.396 and -0.025 in matched SIPP. A larger nominal transition count from overlapping months does not improve the incremental

Table 4. What the reproducible destination-zero endpoint contains

Endpoint or component	Weighted share (%)	Zero rows	CPS incidence reliability
<i>Panel A. CPS annual wage-and-salary income</i>			
Destination annual wage zero, all linked transitions	6.63	11,828	0.940
No work in destination calendar year	64.50		0.840
Worked in destination calendar year	35.50		0.940
Worked all 52 weeks	26.02		0.914
<i>Panel B. SIPP monthly main-job earnings, one transition/person</i>			
Destination monthly main-job zero, all selected transitions	3.51	914	
Self-employed work arrangement	64.47		
Employer work arrangement	34.62		
Any no-job spell recorded in destination month	2.06		

Notes: Overall zero rates use all positive-origin transitions; indented shares condition on a destination zero. Official person weights are used. CPS anatomy rates use 183,809 linked annual transitions across all 72 candidate cells; the reliability entries use unconditional component incidence among 181,946 transitions in the 65 supported cells. Each reliability entry applies the Spearman–Brown transform to the mean independent-person half-sample correlation over 30 splits. The strict SIPP anatomy sample contains 25,310 people and freezes CPS signal information through 2017 before SIPP destination outcomes in 2019–2024. SIPP arrangement and no-job indicators are contemporaneous with the destination month and refer to job 1 rather than total labor income. Arrangement classes are mutually exclusive; the no-job-spell flag can coexist with an arrangement within a month.

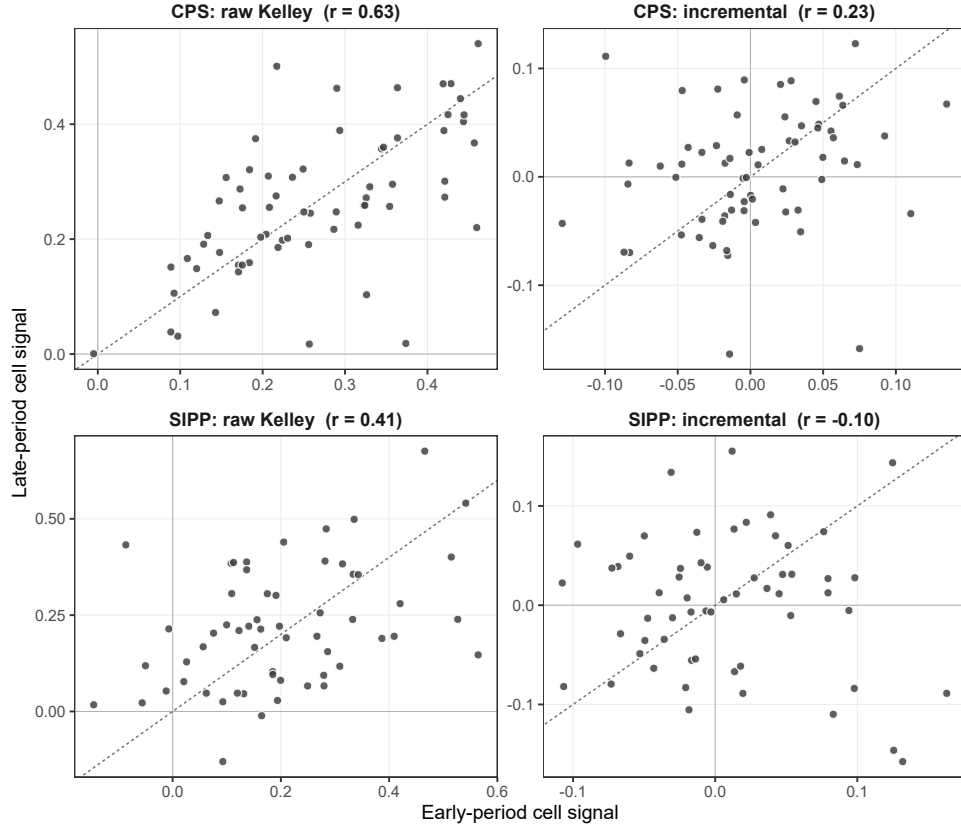
Table 5. Sequential content and time stability of downside Kelley shape

Design	Supported records	Cells	Raw	+Zero/demog.	+Moments	Zero share
CPS pooled exact crosswalk	181,946	65	0.740	0.646	0.370	0.940
CPS early downstream signal	106,404	60	0.687	0.564	0.397	0.909
SIPP one/person	50,137	58	0.602	0.125	0.151	0.809
SIPP repeated months	953,809	60	0.622	0.178	0.070	0.831
<i>Early-versus-late Pearson correlation</i>						
CPS pooled exact crosswalk			0.631	0.591	0.230	0.921
SIPP one/person			0.414	0.153	−0.100	0.691
SIPP repeated months			0.279	0.066	0.054	0.614

Notes: Supported records sum observations in the displayed post-support cells; the underlying candidate samples before cell-support screening contain 183,809 pooled CPS transitions, 108,707 early CPS transitions, 51,152 matched SIPP transitions, and 967,563 overlapping SIPP transitions. Entries above the midrule apply the Spearman–Brown transform to the mean global-person split correlation. “+Zero/demog.” removes the linear cell zero-transition share plus education and age components. “+Moments” also removes quadratic cell mean and standard deviation and is the exact projection used for the portfolio regressor. Pooled CPS uses Kish support of 200 plus 100 in each period; the early row uses Kish support of 200 in the early period and the same 60 cells assigned downstream. SIPP uses effective-person support of 100 pooled and 50 in each period. The matched SIPP transition is drawn once per person before any destination-positive restriction.

ranking. Figure 3 plots the cell-level signals behind these within-survey correlations: the raw ranking retains a visible diagonal in each survey, while the incremental panels are close to exchangeable across periods. Appendix Table B.4 shows the same hierarchy under alternative splits and transition selections, complete-year SIPP main-job earnings, and direct common-year comparison: pooled raw and incremental CPS–SIPP correlations are 0.482 and 0.053.

Figure 3. Within-survey time stability of the cell signal, cell by cell



Notes: Each point is one supported cell; the horizontal axis is the early-period signal and the vertical axis the late-period signal, with the dashed line marking equality and Pearson correlations in the panel titles. Raw Kelley uses the zero-inclusive construction; the incremental signal is the residual from equation (5), re-estimated within each period. CPS panels use the exact crosswalk (65 cells); SIPP panels use one transition per person (58 cells).

5.4 What is and is not measured

The data support two conclusions and rule out a third. First, a survey-specific downside-support environment is measurable: independent people reproduce the zero-inclusive group ranking separately within CPS and SIPP. Second, the stable ranking jointly reflects the zero endpoint, mean growth, dispersion, and whatever shape variation remains. Third, reliability alone cannot apportion that joint object or establish the counterfactual comparison needed for “shape holding the other distributional features fixed.”

Equation (5) is deliberately a conditional diagnostic, not a declaration that correlated variation is spurious. Its lower reliability is also a limit on the portfolio application: attenuation can make the measured incremental coefficient small even when a latent shape contrast matters. The estimator tournament, support frontier, and construction details are reported in Appendix B.

6 Sensitivity to CPS Tail-Measure Estimation

The CPS signal is estimated independently of SIPP, but independence does not make it known. [Ranish \(2013\)](#) provides a direct household-finance precedent for bootstrapping an estimated labor-risk object through a portfolio regression. The primary resampling exercise here draws CPS workers within signal cells, rebuilds the weighted distribution, recomputes support, re-estimates the sequential projection in equation (5), restandardizes the signal, remaps it to both SIPP observations, reconstructs the transition risk sets, and refits the three outcome regressions. Variation across the rebuilt-CPS estimates isolates the component induced by rebuilding the independent measurement sample. I report a normal plug-in sensitivity that adds this CPS component to the conditional two-way-clustered variance. Because SIPP is held fixed, this is not joint inference. It does not correct attenuation or deliver inference for an error-free latent signal. Table 6 reports the complete transition family.

The CPS-first-stage sensitivity separates the transition result from the standing levels. For entry, the across-draw CPS standard deviation is 0.258 point and the plug-in standard error is 0.375 point. The resulting $p = 0.0052$ and three-outcome $q = 0.0157$ remain below 5%; replacing the normal reference with t_{59} gives $p = 0.0070$ and $q = 0.0211$. Exactly 198 of the 200 fixed-SIPP remeasurement coefficients are negative, but their mean is -0.650 point rather than the observed-signal estimate of -1.047 points, a displacement of 0.397 point. Their 2.5th and 97.5th percentiles are -1.125 and -0.138 points. Because SIPP is held fixed and the remeasurement distribution is not centered on the observed estimate, these frequencies and percentiles are diagnostics, not confidence intervals or probability statements. The reported variance sum therefore remains a sensitivity rather than a conventional joint bootstrap. Signed change has plug-in $q = 0.064$, exit remains imprecise, and the standing all-worker share and participation have plug-in $p = 0.109$ and $p = 0.153$.

The tail decomposition uses a second exact-risk-set rebuild. Every draw reconstructs $q_{0.10}$, $q_{0.50}$, $q_{0.90}$, total span, signed composition, and incremental Kelley, while all 27,752 SIPP transitions and 60 assigned cells remain fixed; 59–60 locked cells clear the draw-specific support threshold. The CPS components of the lower- and upper-distance standard errors are 0.562 and 0.619 percentage point, yielding plug-in p -values of 0.0015 and 0.068 and a joint $p = 0.0022$. For the exact span–composition rotation, the CPS components are 1.331 and 0.986 points; signed composition has $p = 0.014$, whereas span has $p = 0.232$. Transforming every draw back to raw arc-growth units rejects equality of the lower and upper marginal effects at $p = 0.0135$. These calculations propagate remeasurement of the assigned distributions within the frozen cell schema; they do not cover assignment-schema choice, deliver joint CPS–SIPP inference, identify household perceptions, or support causal responses.

Alternative centered estimators do not overturn this hierarchy: in the retained broader 2018–2024 tournament, none of the Bowley, moment-skewness, or L-skewness portfolio families passes false-discovery-rate adjustment (minimum $q = 0.244$). The Appendix reports the full estimator grid rather than selecting the closest result.

Table 6. Plug-in transition sensitivity after rebuilding the CPS first stage

Outcome	Est.	Cond. S.E.	Cond. q	CPS S.D.	Plug-in S.E.	Plug-in p	Plug-in q
Signed change	-0.731**	0.260	0.010	0.250	0.360	0.043	0.064
Zero to positive holdings	-1.047***	0.272	0.001	0.258	0.375	0.005	0.016
Positive to zero holdings	-0.813	1.152	0.483	0.845	1.429	0.569	0.569

Notes: Entries are percentage points per one early-CPS cross-cell standard deviation. The calculation uses 200 deterministic CPS person-within-cell resamples and rebuilds weighted quantiles, Kish support selection, zero-share and quadratic mean/dispersion residualization, standardization, assignment to both SIPP observations, and the transition sample. Every draw produces all three estimates; supported cells range from 59 to 61. SIPP is held fixed, while conditional two-way clustering supplies the downstream sampling component. The plug-in calculation is $S.E.^2 = \text{conditional } S.E.^2 + \text{CPS coefficient } S.D.^2$ and uses a two-sided standard-normal reference. It is a sensitivity, not a joint nonparametric bootstrap or inference for an error-free latent signal. Conditional and plug-in q -values each adjust exactly the three displayed transitions. Stars use the displayed unadjusted plug-in p -values: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

The SCE evidence is a separate construct check, not validation of the CPS assignment as perceived risk. Perceived dispersion is associated with a lower standing defined-contribution equity share by 3.63 to 4.73 percentage points across three shape controls (all $p < 0.001$). A centered upside-minus-downside semideviation gap is positively associated with retrospective riskier reallocation ($p = 0.036$), but its primary-family $q = 0.054$, expanded-support $q = 0.169$, null normalized orientation, null standing-share coefficient, and negligible out-of-sample increment keep it exploratory. The separate lower and upper components are correlated 0.95. These respondent-level results neither establish prospective behavioral identification nor show that perceived risk dominates assigned histories (Appendix F).

7 Implications for Measuring and Modeling Labor-Income Risk

The results change what an assigned labor-income-risk measure must establish before it can support a household-finance interpretation. First, the population and support must be named: destination zeros, positive-to-positive conditioning, and the growth transformation define different economic objects. When a quantile sits at the support bound, a tail distance becomes an indicator for that bound, so the share of groups at the bound and the composition of the zeros, here self-employment versus nonwork, belong next to the coefficient. Second, person-level splits should reveal rank reliability and time stability; the number of household observations cannot substitute for the number and precision of measured groups. Third, the conditional interpretation should be explicit, including the statistic's relation to zero incidence, mean, and dispersion. Fourth, inference should carry the estimated distribution, nonlinear quantiles, support selection, residualization, and standardization into the downstream uncertainty calculation.

The household-finance implication goes beyond a measurement checklist. Life-cycle and portfolio models need not force favorable and adverse earnings tails to move through a single variance parameter. The mirror-law exercise in Appendix A.3 supplies only a possibility result: two earnings laws can share the complete distribution of absolute shocks, and hence every even moment, yet

generate different portfolio choices in a one-period problem. It does not map the empirical lower and upper quantile distances into structural shocks or identify the operative mechanism. The estimates instead motivate models in which the two tails may vary separately, with their perceived counterparts measured directly when belief data permit. Appendix Table G.2 gives a compact implementation checklist for assigned-risk designs.

8 Conclusion

This paper shows that favorable and adverse earnings tails contain opposing conditional information about portfolio entry within the baseline public-cell assignment. Among lagged nonholders, a one-standard-deviation wider lower tail predicts a 2.71-percentage-point lower probability of positive reported holdings one year later, while a wider upper tail predicts a 1.92-point higher probability. The pair is jointly distinguishable from zero in the CPS plug-in sensitivity ($p = 0.0022$). In 99,999 cell-level wild-cluster draws, lower and upper distances have adjusted q -values of 0.019 and 0.087 and signed composition has $p = 0.033$; total span remains imprecise in the plug-in calculation, while signed composition has $p = 0.014$. Variance and other one-number scale summaries can therefore conceal information in tail composition. Cross-asset comparisons do not produce sufficiently stable evidence to claim that the pattern is unique to stock-and-mutual-fund entry.

The interpretation is narrower than two robust standalone tail effects. Recomputing the early-CPS tails after excluding destination records classified as self-employed sharply attenuates the lower estimate. Across zero-endpoint exercises, signed composition retains its direction and conditional precision, while the upper estimate remains positive but is imprecise once; under an occupation-based assignment schema, the upper estimate reverses sign. Within-person estimates depend on lagged versus current assignment, and the single pre-pandemic holdout is inconclusive. Historical group distributions remain assigned rather than elicited, selection into risk cells remains endogenous, and the first-stage calculation is a sensitivity rather than joint inference. The evidence therefore describes subsequent positive reported holdings within the baseline assignment, not a causal response, first-ever market entry, or a universal upper-tail effect.

The result gives models intended to confront this evidence a concrete empirical target in which favorable and adverse earnings tails can vary separately rather than being forced through a single variance parameter. The mirror-law benchmark makes only the narrower theoretical point that matching every even moment need not make portfolio choices identical; it neither maps the empirical quantile gaps into structural shocks nor identifies whether the association operates through beliefs, participation costs, cash flow, or sorting. A natural next step is to measure perceived lower and upper tails prospectively and connect them to verified portfolio flows.

Data Availability. The CPS ASEC extract uses licensed IPUMS CPS microdata (Flood et al., 2025). SIPP (U.S. Census Bureau, 2025), SCE (Federal Reserve Bank of New York, 2026), and SCF (Board of Governors of the Federal Reserve System, 2019) are public; the replication package supplies download links, build scripts, frozen analysis samples, and source-to-paper validation.

References

- Akey, Pat, Adriana Z. Robertson, and Mikhail Simutin, 2026, Noisy factors? The retroactive impact of methodological changes on the Fama–French factors, *Review of Finance*, rfag002, doi:10.1093/rof/rfag002.
- Angerer, Xiaohong, and Pok-Sang Lam, 2009, Income risk and portfolio choice: An empirical study, *Journal of Finance* 64, 1037–1055.
- Arellano, Manuel, Richard Blundell, and Stéphane Bonhomme, 2017, Earnings and consumption dynamics: A nonlinear panel data framework, *Econometrica* 85, 693–734.
- Arellano, Manuel, Orazio Attanasio, Margherita Borella, Mariacristina De Nardi, and Gonzalo Paz-Pardo, 2026, Subjective earnings and employment dynamics, NBER Working Paper No. 35027, doi:10.3386/w35027.
- Benjamini, Yoav, and Yosef Hochberg, 1995, Controlling the false discovery rate: A practical and powerful approach to multiple testing, *Journal of the Royal Statistical Society: Series B (Methodological)* 57, 289–300.
- Betermier, Sébastien, Thomas Jansson, Christine Parlour, and Johan Walden, 2012, Hedging labor income risk, *Journal of Financial Economics* 105, 622–639.
- [dataset] Board of Governors of the Federal Reserve System, 2019, 2019 Survey of Consumer Finances public data, Washington, DC, <https://www.federalreserve.gov/econres/scfindex.htm>.
- Brown, William, 1910, Some experimental results in the correlation of mental abilities, *British Journal of Psychology* 3, 296–322.
- Busch, Christopher, David Domeij, Fatih Guvenen, and Rocio Madera, 2022, Skewed idiosyncratic income risk over the business cycle: Sources and insurance, *American Economic Journal: Macroeconomics* 14, 207–242.
- Cameron, A. Colin, Jonah B. Gelbach, and Douglas L. Miller, 2011, Robust inference with multiway clustering, *Journal of Business & Economic Statistics* 29, 238–249.
- Caplin, Andrew, Victoria Gregory, Eungik Lee, Søren Leth-Petersen, and Johan Sæverud, 2026, Subjective earnings risk, *Review of Economic Studies*, rdag070, doi:10.1093/restud/rdag070.
- Catherine, Sylvain, 2022, Countercyclical labor income risk and portfolio choices over the life cycle, *Review of Financial Studies* 35, 4016–4054.
- Catherine, Sylvain, Paolo Sodini, and Yapei Zhang, 2024, Countercyclical income risk and portfolio choices: Evidence from Sweden, *Journal of Finance* 79, 1755–1788.
- Chang, Yongsung, Jay H. Hong, and Marios Karabarbounis, 2018, Labor market uncertainty and portfolio choice puzzles, *American Economic Journal: Macroeconomics* 10, 222–262.
- Chang, Yongsung, Jay H. Hong, Marios Karabarbounis, Yicheng Wang, and Tao Zhang, 2022, Income volatility and portfolio choices, *Review of Economic Dynamics* 44, 65–90.
- Cocco, João F., Francisco J. Gomes, and Pascal J. Maenhout, 2005, Consumption and portfolio choice over the life cycle, *Review of Financial Studies* 18, 491–533.
- d’Astous, Philippe, and Stephen H. Shore, 2024, Human capital risk and portfolio choices: Evidence from university admission discontinuities, *Journal of Financial Economics* 154, 103793.

- Dew-Becker, Ian, 2024, Real-time forward-looking skewness over the business cycle, *Review of Economic Dynamics* 54, 101233, doi:10.1016/j.red.2024.101233.
- Drerup, Tilman H., Matthias Wibrall, and Christian Zimpelmann, 2023, Skewness expectations and portfolio choice, *Experimental Economics* 26, 107–144, doi:10.1007/s10683-022-09780-9.
- Fagereng, Andreas, Charles Gottlieb, and Luigi Guiso, 2017, Asset market participation and portfolio choice over the life-cycle, *Journal of Finance* 72, 705–750, doi:10.1111/jofi.12484.
- Fagereng, Andreas, Luigi Guiso, and Luigi Pistaferri, 2018, Portfolio choices, firm shocks, and uninsurable wage risk, *Review of Economic Studies* 85, 437–474.
- [dataset] Federal Reserve Bank of New York, 2026, Survey of Consumer Expectations public data bank, New York, NY, <https://www.newyorkfed.org/microeconomics/databank>.
- [dataset] Flood, Sarah, Miriam King, Renae Rodgers, Steven Ruggles, J. Robert Warren, Daniel Backman, Etienne Breton, Grace Cooper, Julia A. Rivera Drew, Stephanie Richards, David Van Riper, and Kari C. W. Williams, 2025, IPUMS CPS: Version 13.0, IPUMS, Minneapolis, MN, doi:10.18128/D030.V13.0.
- Gálvez, Julio, and Gonzalo Paz-Pardo, 2026, Richer earnings dynamics, consumption and portfolio choice over the life cycle, *Journal of Financial Economics* 176, 104206, doi:10.1016/j.jfineco.2025.104206.
- Giglio, Stefano, Matteo Maggiori, Johannes Stroebel, and Stephen Utkus, 2021, Five facts about beliefs and portfolios, *American Economic Review* 111, 1481–1522, doi:10.1257/aer.20200243.
- Gomes, Francisco, and Alexander Michaelides, 2005, Optimal life-cycle asset allocation: Understanding the empirical evidence, *Journal of Finance* 60, 869–904, doi:10.1111/j.1540-6261.2005.00749.x.
- Guiso, Luigi, Tullio Jappelli, and Daniele Terlizzese, 1996, Income risk, borrowing constraints, and portfolio choice, *American Economic Review* 86, 158–172.
- Guvenen, Fatih, Fatih Karahan, Serdar Ozkan, and Jae Song, 2021, What do data on millions of U.S. workers reveal about lifecycle earnings dynamics?, *Econometrica* 89, 2303–2339.
- Guvenen, Fatih, Serdar Ozkan, and Jae Song, 2014, The nature of countercyclical income risk, *Journal of Political Economy* 122, 621–660.
- Harmenberg, Karl, 2021, The labor-market origins of cyclical skewness, Working paper, Copenhagen Business School.
- Hosking, J. R. M., 1990, L-moments: Analysis and estimation of distributions using linear combinations of order statistics, *Journal of the Royal Statistical Society: Series B (Methodological)* 52, 105–124.
- Kim, Tae-Hwan, and Halbert White, 2004, On more robust estimation of skewness and kurtosis, *Finance Research Letters* 1, 56–73.
- Kish, Leslie, 1965, *Survey Sampling* (Wiley, New York, NY).
- Moulton, Brent R., 1990, An illustration of a pitfall in estimating the effects of aggregate variables on micro units, *Review of Economics and Statistics* 72, 334–338.

- Murphy, Kevin M., and Robert H. Topel, 1985, Estimation and inference in two-step econometric models, *Journal of Business & Economic Statistics* 3, 370–379.
- Pagan, Adrian, 1984, Econometric issues in the analysis of regressions with generated regressors, *International Economic Review* 25, 221–247.
- Ranish, Benjamin, 2013, Why do households with risky labor income take greater financial risks?, Available at SSRN 2681360, doi:10.2139/ssrn.2681360.
- Roodman, David, James G. MacKinnon, Morten Ørregaard Nielsen, and Matthew D. Webb, 2019, Fast and wild: Bootstrap inference in Stata using boottest, *Stata Journal* 19, 4–60, doi:10.1177/1536867X19830877.
- Shen, Jialu, 2024, Countercyclical risks, consumption, and portfolio choice: Theory and evidence, *Management Science* 70, 2862–2881.
- Spearman, Charles, 1910, Correlation calculated from faulty data, *British Journal of Psychology* 3, 271–295.
- Törnqvist, Leo, Pentti Vartia, and Yrjö O. Vartia, 1985, How should relative changes be measured?, *The American Statistician* 39, 43–46.
- [dataset] U.S. Census Bureau, 2025, Survey of Income and Program Participation public-use data, 2014–2023 panels, Washington, DC, <https://www.census.gov/programs-surveys/sipp/data/datasets.html>.
- van den Akker, Ruben, Jochem de Bresser, Federica Teppa, and Serena Trucchi, 2026, Subjective expectations and household portfolio choices, Work in progress presented at the 2026 IAAE Annual Conference, <https://editorialexpress.com/conference/IAAE2026/program/IAAE2026.html>.
- Viceira, Luis M., 2001, Optimal portfolio choice for long-horizon investors with nontradable labor income, *Journal of Finance* 56, 433–470.
- Vissing-Jorgensen, Annette, 2002, Towards an explanation of household portfolio choice heterogeneity: Nonfinancial income and participation cost structures, NBER Working Paper 8884, doi:10.3386/w8884.
- Wang, Tao, 2023, Perceived versus calibrated income risks in heterogeneous-agent consumption models, Bank of Canada Staff Working Paper 2023-59, doi:10.34989/swp-2023-59.

Online Appendix
for
**Beyond Variance:
Asymmetric Labor Income Risk and Portfolio Entry**

Contents

A	Estimator definitions and orientation	A.2
B	Reliability design and additional results	A.4
C	CPS-to-SIPP crosswalk and portfolio construction	A.7
D	Identification and construct checks for the entry result	A.13
E	Portfolio parameterizations	A.15
F	Subjective-belief construct and timing checks	A.16
G	Plug-in first-stage uncertainty	A.20
H	Scope of the group-generated-risk workflow	A.21

A Estimator Definitions and Orientation

Let q_p denote a survey-weighted quantile of growth g , and let μ and σ denote its survey-weighted mean and standard deviation. Every estimator is oriented so that a positive value denotes a more adverse downside distribution.

Table A.1. Distributional estimators

Estimator	Definition	Interpretation
Upper tail	$U = q_{.90} - q_{.50}$	Magnitude above the median
Lower tail	$L = q_{.50} - q_{.10}$	Magnitude below the median
90–10 spread	$S = L + U = q_{.90} - q_{.10}$	Interquantile scale
Tail difference	$D = L - U$	Unscaled downside-minus-upside length
Kelley shape	$(L - U)/(L + U)$	Tail difference normalized by the 90–10 spread
Bowley shape	$[(q_{.50} - q_{.25}) - (q_{.75} - q_{.50})]/(q_{.75} - q_{.25})$	Central-quantile analogue of Kelley shape
Moment skewness	$-\mathbb{E}[(g - \mu)/\sigma]^3$	Conventional skewness with sign reversed
Tail probability difference	$\Pr(g \leq \mu - \sigma) - \Pr(g \geq \mu + \sigma)$	Downside-minus-upside mass beyond one SD
Conditional tail balance	Tail probability difference divided by its probability sum	Direction conditional on being in either tail
L-skewness	$-\lambda_3/\lambda_2$ from probability-weighted L-moments (Hosking, 1990)	Robust full-distribution orientation
Incremental estimator	Residual from equation (5), applied to the named estimator	Cell ranking incremental to zero share, location, and scale

Weighted quantiles sort observations by g , cumulate survey weights, and select the first value whose cumulative weight share meets the requested probability. L-moments are approximated on a probability grid from 0.005 through 0.995. Weighted variance divides by the sum of weights because the target is the weighted empirical distribution in the cell.

For Kelley shape, $K = 1$ is the limiting case in which the 90–10 span lies entirely below the median, while $K = -1$ is the limiting upside case. A point mass at $g = -2$ can move the lower quantile to the support bound. This changes the unconditional distribution’s shape; it is not treated as an outlier correction.

A.1 Proofs of the measurement lemmas

Proof of Lemma 1. Because a lies below the support of F , the mixture distribution is zero below a , jumps to p at a , and equals $p + (1 - p)F(x)$ above a . The generalized inverse is therefore a for $\tau \leq p$. For $\tau > p$, solving $p + (1 - p)F(x) \geq \tau$ gives the stated transformed quantile of F .

The mean follows by iterated expectations. Conditioning on the mixture component and applying the law of total variance gives

$$\text{Var}(G_p) = (1 - p)\sigma_F^2 + p(1 - p)(\mu_F - a)^2.$$

Changing p therefore changes the quantile map and lower moments while leaving the conditional distribution F untouched. The formula imposes no global monotonicity on Kelley shape without additional restrictions on F . $\square$

Proof of Lemma 2. Independence implies that the covariance between the two half-sample estimates is $\text{Var}(a'Z_c + s_c)$. Each half-sample variance is $\text{Var}(a'Z_c + s_c) + \sigma_e^2$. Dividing covariance by the common variance gives the displayed correlation. Setting $\text{Var}(s_c) = 0$ and taking σ_e^2 to zero establishes the final statement. $\square$

A.2 Why arc growth is the baseline

For nonnegative earnings y_0 and y_1 , with $y_0 > 0$, arc growth is defined at $y_1 = 0$ and equals -2 . Writing $y_1 = y_0(1 + x)$,

$$g^{\text{arc}} = \frac{2x}{2+x} = x + O(x^2), \quad \log(y_1/y_0) = x + O(x^2).$$

Unlike $\log(y+1)$ growth, it is invariant to the reporting unit. Unlike positive-to-positive log growth, it retains a transition out of positive earnings. Its bounded support and lower-bound mass are disclosed features of the estimand.

A.3 A qualitative orientation benchmark

This benchmark separates a valid local third-moment comparative static from the empirical quantile-tail coefficients. Let x denote terminal financial wealth before labor income and define

$$y = 1 + \sigma_y \eta, \quad \mathbb{E}\eta = 0, \quad \mathbb{E}\eta^2 = 1, \quad m_3 = \mathbb{E}\eta^3,$$

with $x + y > 0$ almost surely. For CRRA utility, let

$$V(x; m_3) = \mathbb{E}u(x + y), \quad A_V(x; m_3) = -\frac{V''(x; m_3)}{V'(x; m_3)}.$$

Within a smooth local family of shock laws that holds the first two moments fixed while varying m_3 , suppose consumption remains bounded away from zero locally and derivatives of the higher-moment remainder are regular. A third-order expansion around $\sigma_y = 0$ gives

$$\frac{\partial A_V(x; m_3)}{\partial m_3} = -\frac{1}{2}\gamma(\gamma+1)(\gamma+2)\sigma_y^3(x+1)^{-4} + O(\sigma_y^4).$$

The leading term is negative; under the stated regularity conditions, the derivative is therefore negative for sufficiently small σ_y . Thus, locally and within this approximation, a higher third central moment lowers derived absolute risk aversion.

For an interior unconstrained risky position in the local mean–variance approximation, write terminal financial wealth as $x = wR_f + w\alpha(R - R_f)$, with the risky return independent of labor income. A second-order expansion around wR_f gives the approximate terminal-wealth certainty-equivalent gain

$$G(w; m_3) \simeq \frac{(\mathbb{E}R - R_f)^2}{2A_V(wR_f; m_3) \text{Var}(R)}.$$

The preceding comparative static raises this approximate gain locally. If a fixed cost creates a regular participation boundary, households sufficiently close to that boundary can change participation. This is not a global threshold theorem: portfolio constraints, a discrete wealth grid, and the cost itself can create flat regions and jumps.

The calculation does not sign the effect of the empirical lower or upper quantile distance. Holding mean and variance fixed, widening $q_{0.50} - q_{0.10}$ need not lower m_3 , and widening $q_{0.90} - q_{0.50}$ need not raise it; either transformation can move the opposite tail and higher moments. The calculation is therefore a qualitative third-moment mechanism, not a prediction for the regression coefficients or their magnitudes.

A separate mirror comparison isolates a stronger distributional contrast. Let F be a centered, unit-variance law formed from CPS arc-growth residuals and let F^- be the law of the negative of every realization. The two laws have the same distribution of absolute shocks and therefore identical even moments, while their odd moments have opposite signs. In the reported numerical one-period CRRA grid, replacing F with F^- changes the optimal risky position in some states. This establishes that matching every even moment need not make portfolio choice invariant. The exercise is not fitted to the SIPP entry rate and is not used to validate the empirical signs, magnitudes, or operative mechanism.

B Reliability Design and Additional Results

B.1 Person-level splitting

Each repetition draws one half indicator for every unique person, attaches it to all records of that person, rebuilds weighted cell distributions independently in each half, re-estimates any projection separately within each half, and correlates the common eligible cells. Eligibility is frozen from the full sample; each half must also clear its fixed support floor. The same split assignment is used across paired construction comparisons.

The main sequential audit uses 30 deterministic person splits. The alternative-estimator audit uses 50. A row split is not valid in SIPP because one worker can contribute repeated monthly year-apart transitions; the matched audit instead draws one eligible transition per person before applying destination-positive restrictions.

Table B.1. Positive-to-positive and zero-inclusive Kelley reliability

Data design	Growth construction	Cells	Raw reliability
CPS age-refined cells	Arc, zeros retained	60	0.735
CPS age-refined cells	Arc, positive-to-positive	60	0.018
CPS age-refined cells	Log, positive-to-positive	60	0.051
SIPP one transition/person	Arc, zeros retained	58	0.602
SIPP one transition/person	Log($y + 1$), zeros retained	58	0.630
SIPP one transition/person	Arc, positive-to-positive	58	0.147
SIPP one transition/person	Log, positive-to-positive	58	0.120

Entries are Spearman–Brown transforms of mean independent-person half-sample correlations. The CPS construction audit uses the corrected IPUMS education mapping. The SIPP comparison selects one year-apart transition per person before construction-specific destination restrictions and uses identical split seeds in every row.

B.2 Population-support construction

B.3 Support frontier

Table B.2. Selected CPS reliability-frontier values

Effective support floor	Common cells	90–10 spread	Raw Kelley	Mean/SD residual Kelley
100	63	0.922	0.728	0.379
200	60	0.929	0.746	0.334
400	58	0.921	0.741	0.368
800	51	0.937	0.791	0.297
1,200	34	0.951	0.839	0.344

General CPS construction tournament, 50 global person splits. This robustness frontier predates the sequential zero-share projection and therefore labels its final column explicitly as the mean/SD residual. Main Table 5 uses the exact portfolio crosswalk and the full sequential projection.

At support 400, early-versus-late Pearson correlations in the broader CPS tournament are 0.731 for the upper tail, 0.794 for the lower tail, 0.840 for the spread, 0.603 for raw Kelley, and 0.109 for mean/SD-residual Kelley. Rank correlations are 0.648, 0.785, 0.790, 0.622, and 0.225. Time stability and split reliability are distinct: the former changes both people and calendar periods, while the latter changes people holding the period fixed.

B.4 Alternative centered estimators

For Bowley, moment, and L-skewness, the raw-controlled and residualized parameterizations' p -values agree to numerical tolerance, as required by Frisch–Waugh–Lovell. None of the twelve alternative-estimator portfolio coefficients survives within-estimator family adjustment. The negative Bowley reliability is not interpreted as a negative population reliability; it indicates that no stable incremental component is detected.

Table B.3. Estimator robustness in the corrected exact crosswalk

Estimator	Raw rel.	Incr. rel.	Raw early/late	Incr. early/late	Min. 2018–24 BH q
Kelley	0.740	0.370	0.631	0.230	0.058
Bowley	0.768	−0.051	0.625	0.249	0.824
Moment skewness	0.794	0.233	0.733	0.290	0.244
L-skewness	0.685	0.178	0.597	0.050	0.381

Reliability is the Spearman–Brown transform of the mean global-person split correlation. Incremental signals remove the linear zero share, quadratic mean and standard deviation, and education and age components, re-estimated inside every split. Early/late entries are Pearson correlations. The last column retains the broader 2018–2024 estimator tournament and is the smallest conditional false-discovery-rate q across participation, positive-financial-assets share, all-worker share, and holder share. The Kelley minimum in that broader window does not pass a 5% family-adjusted threshold and its corresponding positive-assets plug-in first-stage-uncertainty p -value is 0.115. The strict-post Kelley family appears in the main text; alternative estimators were not selected for a new strict-window tournament. The replication output contains the complete estimator-by-outcome coefficient grid.

B.5 Assignment, transition-selection, and cross-survey sensitivities

Table B.4 separates sensitivity to analyst-generated person splits and transition selections from formal sampling uncertainty. It also replaces the overlapping-month SIPP object with complete calendar-year main-job earnings and directly compares CPS and SIPP cells over common origin years.

Table B.4. Reliability and cross-survey sensitivity ranges

Design	Raw Kelley	Incremental Kelley	Zero share
<i>Panel A. Spearman–Brown reliability: center [P10, P90]</i>			
CPS exact crosswalk, 30 person splits	0.740 [0.679, 0.802]	0.370 [0.109, 0.549]	0.940 [0.916, 0.956]
SIPP, 30 one-transition selections	0.566 [0.501, 0.630]	0.092 [−0.072, 0.239]	0.829 [0.814, 0.851]
SIPP, complete-year main-job earnings	0.442 [0.213, 0.607]	0.128 [−0.203, 0.369]	0.774 [0.703, 0.859]
<i>Panel B. Direct common-year CPS–SIPP Pearson correlation: mean [P10, P90]</i>			
Pooled common-year cells	0.482 [0.440, 0.522]	0.053 [−0.025, 0.137]	0.619 [0.598, 0.633]

Notes: Brackets are analyst-generated sensitivity ranges, not confidence intervals. The CPS row varies the 30 frozen person-half assignments; its center applies the Spearman–Brown transform to the mean half-sample correlation. The second row reports the median and P10–P90 range across 30 deterministic choices of one monthly 12-month-apart transition per person, with 30 person splits per choice. The complete-year row sums twelve finite nonnegative decoded monthly main-job earnings observations, uses December origin characteristics, selects one adjacent-year transition per person, and reports the range across 30 person splits; it contains 40,635 people in 57 supported cells. Panel B compares 58 common cells over the eight origin years shared by both surveys and varies the SIPP transition selection 30 times. CPS measures total annual wage-and-salary earnings, whereas SIPP measures main-job earnings; the direct correlation is therefore a comparability sensitivity, not validation against an identical construct. Incremental Kelley is residualized on the zero share, quadratic mean and standard deviation, education, and age within each design.

Table B.5. Strict out-of-time prediction across SIPP transition selections

Later SIPP destination outcome	Early CPS signal	Median [P10, P90]	Positive / $q < 0.05$
Any monthly main-job zero	Raw Kelley	0.297 [0.068, 0.450]	29 / 0
Any monthly main-job zero	Incremental Kelley	0.154 [−0.013, 0.338]	26 / 0
Any monthly main-job zero	Destination-zero share	0.965 [0.104, 1.662]	28 / 0
Employer-arrangement zero	Raw Kelley	0.200 [0.081, 0.328]	30 / 1
Self-employment-arrangement zero	Raw Kelley	0.059 [−0.045, 0.165]	22 / 0
Zero with any no-job spell	Raw Kelley	−0.006 [−0.043, 0.057]	14 / 0

Entries are percentage-point coefficients per one early-CPS signal standard deviation across 30 deterministic choices of one eligible 2018–2023 SIPP origin transition per person, selected without conditioning on destination state. Destination outcomes occur in 2019–2024, after the final destination year entering the CPS signal. Each cell-year regression uses 360 cell-years in 60 signal cells, official person weights, baseline earnings and demographics, destination-year and additive industry, education, and age-band fixed effects, and standard errors clustered by signal cell. The incremental specification also includes the CPS zero share and quadratic mean and dispersion. The final column counts positive coefficients and selections whose four-outcome Benjamini–Hochberg q -value is below 0.05. Brackets are transition-selection sensitivity ranges, not sampling confidence intervals. SIPP arrangement classes are mutually exclusive; the no-job-spell flag can coexist with an arrangement within a month.

C CPS-to-SIPP Crosswalk and Portfolio Construction

C.1 Exact public industry mapping

Table C.1. CPS IND1990 to SIPP broad-industry crosswalk

SIPP code	Sector	CPS IND1990 support
1	Agriculture	10–32
2	Mining	40–50
3	Construction	60
4	Manufacturing	100–392
5	Trade	500–691
6	Transportation and utilities	400–432
7	Information	440–472
8	Finance, insurance, and real estate	700–712
9	Professional and business services	721–760 and 861–893
10	Education and health	800–860
11	Leisure and hospitality	761–791
12	Other services	No clean IND1990 analogue; unmatched
13	Public administration	900–932
14	Military	940–960; excluded ex ante

Education is harmonized to high school or less, some college, and college; IPUMS CPS code 73 is a high-school diploma and belongs to the first group. Age is divided into 25–44 and 45–65. The 12 matched industries, three education groups, and two age bands give 72 candidate cells. Base support requires pooled CPS Kish support of at least 200 and at least 100 in each early and late period; 65 cells pass. The early-signal rule requires early CPS Kish support of 200; 60 cells pass.

The SIPP source contains 7,026,846 person-months on 263,280 identifiers. Age, weight, and industry restrictions leave 2,495,083 worker-months on 109,401 people. Annualization produces 224,910 worker-years on 108,606 people before the corrected CPS support merge. The pooled supported merge contains 212,635 worker-years on 104,093 people. The strict-post primary sample contains 90,110 worker-years on 52,115 people during 2019–2024 in 60 signal cells; every outcome therefore postdates the CPS destination years, which end in 2018. The broader 2018–2024 timing-sensitivity window contains 107,555 worker-years on 61,162 people in the same 60 cells.

C.2 Portfolio outcomes

Let M_{it} denote the observed stock-and-mutual-fund balance and F_{it} reported financial assets. The level outcomes are

$$\text{PART}_{it} = \mathbf{1}\{M_{it} > 0\}, \quad (\text{C.1})$$

$$\text{SMF}_{it}^+ = 100M_{it}/F_{it} \quad \text{for } F_{it} > 0, \quad (\text{C.2})$$

$$\text{SMF}_{it}^{\text{all}} = 100M_{it}/F_{it} \quad \text{when } F_{it} > 0, \text{ and } 0 \text{ otherwise}, \quad (\text{C.3})$$

$$\text{SMF}_{it}^{\text{holder}} = 100M_{it}/F_{it} \quad \text{for } M_{it} > 0, F_{it} > 0. \quad (\text{C.4})$$

The ratio is not total household equity exposure. Mutual funds need not be pure equity, retirement balances enter reported financial assets without their internal allocation being observed, and employer-stock compensation cannot be cleanly removed.

C.3 Portfolio sample and outcome audit

The person-year construction averages the available monthly balance-sheet records within a calendar year. Table C.2 documents the strict-post early-CPS-to-later-SIPP frame before any outcome-specific deletion and retains the broader-window count for comparison. The cached cleaned file contains no asset-imputation flags, so source imputations cannot be classified; missingness, denominator support, ratio bounds, and repeated records can nevertheless be audited directly.

Table C.2. SIPP portfolio-sample audit

Audit item	Value	Interpretation
Strict-post primary frame	90,110	Worker-years on 52,115 people during 2019–2024 in 60 supported cells
Broader timing-sensitivity frame	107,555	Worker-years on 61,162 people during 2018–2024 in the same cells
Strict-post positive financial assets	93.40%	Observations entering the positive-assets share
Strict-post weighted participation	19.49%	Positive observed stock-and-mutual-fund balance
Strict-post all-worker share equal to zero	80.49%	Nonholders, including nonpositive financial-asset years
Strict-post fewer than 12 eligible monthly records	15.77%	Annual mean uses the available within-year records
Strict-post positive-assets ratio outside [0, 100]	0.00%	No mechanical ratio-tail deletion is required
Broader-frame person-years in multiworker household-years	61.09%	More than one eligible worker shares the household-year
Broader-frame multiworker household-years with same asset tuple	0.87%	Exact cross-worker duplication is rare

Values are computed before outcome-specific missing-value deletion. Strict-post outcome diagnostics apply the 2019–2024 restriction to the frozen worker-year file. The two explicitly labeled household-record rows come from the broader 2018–2024 raw-record audit. The asset tuple comprises the annual stock-and-mutual-fund balance, financial assets, total assets, and net worth. Monthly asset values are constant within each retained person-year in the cached source. Official person-year weights are used only for the participation rate; the remaining shares are unweighted construction diagnostics. Machine-readable broader-window definitions and weighted ratio quantiles are produced by `15_portfolio_sample_audit.R`; the timing restriction and exact strict-post counts are reproduced by `17_post2019_portfolio_timing.R`.

The strict-post frame maps 90,110 worker-years into 61,658 household-years and 36,333 distinct households, using the public SIPP composite household key SSUID plus SHHADID. SSUID alone would merge multiple current households in 811 sample-unit-years, representing 1,651 distinct household-years. Of the correctly keyed household-years, 36,049 contain exactly one worker who satisfies the age, industry, education, supported-cell, and regression-sample rules. Table C.3 therefore changes the inference cluster in the full frame and then removes household-years with multiple supported analytic workers. Neither exercise identifies the household’s financial decision maker, but together they test whether repeated workers attached to the same household drive the conditional level results.

Table C.3. Household decision-unit sensitivity, strict-post sample

Outcome	Raw Kelley	Raw q	Incremental Kelley	Incremental q	N
<i>Panel A. Full worker frame; household and signal-cell clustering</i>					
Participation	0.0038 (0.0073)	0.879	−0.0092* (0.0048)	0.080	90,110
Share, positive financial assets	0.0608 (0.1836)	0.879	−0.2610*** (0.0976)	0.039	84,161
Share, all workers	0.0800 (0.1833)	0.879	−0.2341** (0.0985)	0.042	90,110
Share, current holders	−0.0967 (0.6303)	0.879	−0.1525 (0.4292)	0.724	17,578
<i>Panel B. One supported analytic worker per household-year</i>					
Participation	0.0053 (0.0086)	0.538	−0.0108* (0.0058)	0.091	36,049
Share, positive financial assets	0.2704 (0.2272)	0.344	−0.3123** (0.1501)	0.091	33,228
Share, all workers	0.2648 (0.2189)	0.344	−0.2742* (0.1382)	0.091	36,049
Share, current holders	0.9947 (0.8708)	0.344	−0.2065 (0.8321)	0.805	6,497

Coefficients are per one early-CPS signal standard deviation; share outcomes are percentage points. Standard errors in parentheses cluster by the composite SIPP household key (SSUID plus SHHADID) and early-signal cell. Superscripts use unadjusted two-sided conditional tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. Benjamini–Hochberg q -values are computed separately across the four outcomes within each panel and signal specification. Panel B retains a household-year only when exactly one worker appears in the supported strict-post analytic frame; this does not assert that the retained person is the household’s sole earner or financial decision maker. The selected person’s official weight is retained because the public cache contains no dedicated household weight. These conditional estimates do not propagate CPS first-stage uncertainty.

The full-frame all-worker coefficient is unchanged by construction and remains -0.234 ; household rather than person clustering gives an SE of 0.099 and $q = 0.042$. In the single-supported-worker restriction, the corresponding estimate is -0.274 (SE 0.138; $q = 0.091$). Thus the sign and magnitude do not depend on counting multiple supported workers in a household-year, although the narrower sample loses precision and remains a conditional generated-regressor sensitivity rather than causal or household-representative evidence.

C.4 Transition durability and positive-balance thresholds

Table C.4 asks whether the nonholder-to-positive-holdings result is a one-year classification or a token-balance artifact. Sustained entry requires zero holdings at $t - 1$ and positive holdings at both t and $t + 1$. The two threshold outcomes instead require a nominal stock-and-mutual-fund balance to move from no more than \$1,000 or \$5,000 to above that threshold. Each outcome is estimated in the primary frame and in the same-signal-cell restriction.

Table C.4. Durability and positive-balance threshold checks

Outcome	Design	Coef.	S.E.	p	BH q	N
Sustained entry through $t + 1$	Primary	−0.697**	0.293	0.0205	0.041	9,331
Sustained entry through $t + 1$	Same cell	−0.578**	0.268	0.0349	0.045	7,328
Cross from $\leq \$1,000$ to $> \$1,000$	Primary	−1.104***	0.222	< 0.0001	< 0.001	28,686
Cross from $\leq \$1,000$ to $> \$1,000$	Same cell	−1.131***	0.273	0.0001	< 0.001	22,242
Cross from $\leq \$5,000$ to $> \$5,000$	Primary	−0.453**	0.221	0.0445	0.045	29,823
Cross from $\leq \$5,000$ to $> \$5,000$	Same cell	−0.537**	0.260	0.0436	0.045	23,132

Coefficients and standard errors are percentage points per one early-CPS incremental-signal standard deviation. Specifications retain the lagged signal, lagged controls, current-year weight, fixed effects, and two-way person and lag-cell clustering of the primary transition design. Superscripts use unadjusted conditional two-sided tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. The BH column adjusts once across all six displayed rows. These are conditional generated-regressor checks and do not add CPS first-stage uncertainty. Thresholds are nominal and are not inflation-adjusted. Sustained-entry samples end in 2023 so that $t + 1$ is observed.

The primary entry estimate is also negative in every leave-one-outcome-year regression, from −1.203 to −0.881 point; the largest conditional p -value is 0.0031. In a stable-household transition frame, the estimate is −1.057 point (SE 0.275; three-outcome $q = 0.001$). Selecting the highest-earning eligible worker in each household transition gives −0.869 point (SE 0.360; $q = 0.056$). The latter narrowly misses 5% after its three-outcome adjustment, so these rows support decision-unit stability rather than a household-pooling mechanism.

Table C.5. Finite-cell inference for the entry tail decomposition

Component	Coef.	Two-way p	Wild p	Wild BH q
Lower distance, $q_{0.50} - q_{0.10}$	−2.712***	< 0.001	0.0097	0.0194
Upper distance, $q_{0.90} - q_{0.50}$	1.917**	0.0272	0.0865	0.0865
Downside composition, $L - U$	−4.456***	0.0051	0.0327	—

Coefficients are percentage points per one early-CPS cross-cell standard deviation in the locked 27,752-transition entry risk set. The conventional column uses two-way person and signal-cell clustering. The wild column uses 99,999 null-imposed Rademacher wild-bootstrap draws with weights at the 60 CPS signal cells and retains the two-way covariance structure (Roodman et al., 2019). The BH family contains the lower and upper coefficients; signed composition is an exact rotation of the same two-dimensional column space and is reported separately. Superscripts use the conventional two-way-clustered tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. This check conditions on the frozen CPS tail estimates; Table 1 separately reports CPS first-stage uncertainty.

Table C.6. Formal contrasts with other asset-entry margins

Comparison entry	Lower contrast	Lower BH q	Upper contrast	Upper BH q	Joint p / BH q	N
Bank deposits	−3.038	0.689	4.717	0.689	0.585 / 0.750	2,712
Bonds	−2.512***	0.086	2.013*	0.279	0.0035 / 0.017	26,913
Retirement assets	−3.033	0.324	3.566	0.324	0.110 / 0.274	12,312
Home equity	−0.485	0.821	0.310	0.838	0.920 / 0.920	13,137
Vehicle equity	−2.265	0.689	2.356	0.689	0.600 / 0.750	7,864
Bonds, one worker per lag household	−2.253**	0.234	2.492**	0.234	0.020 / 0.102	20,337

Each row restricts the locked worker-transition frame to records whose household reports neither stock-and-mutual-fund holdings nor the comparison asset in the lag year; N counts worker transitions. The dependent variable is stock-and-mutual-fund entry minus comparison-asset entry, so a coefficient is a formal cross-outcome slope difference on the same observations. Coefficients are percentage points per one early-CPS cross-cell standard deviation. Plug-in covariance adds the covariance of coefficients across CPS person-within-cell resamples to the locked two-way-clustered SIPP covariance. Stars use the unadjusted plug-in two-sided tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. Lower and upper BH q -values adjust one family of ten scalar contrasts; joint q -values adjust the five two-dimensional tests. The first five rows retain all eligible workers, cluster by person and signal cell, and use 200 CPS draws. Because the asset reports are household-level, the final row re-estimates the complete five-comparison family with 1,000 CPS draws after selecting the highest-lag-income eligible worker in each lag-household transition and clusters by lag household and signal cell; only its bond row is displayed. The all-worker bond vector passes its joint family, but the one-worker version narrowly misses the 10% joint-family threshold and neither of its scalar contrasts survives the ten-coefficient family. These are exploratory diagnostics, not literal placebos: labor-income tails can affect other assets for economic reasons, and neither a rejection nor a null contrast establishes a stock-and-mutual-fund mechanism.

Table C.7. Post hoc entry influence and specification stress tests

Diagnostic	Coef.	S.E.	Conditional p	N
Lag signal-cell by outcome-year aggregation	−0.985***	0.297	0.0016	300
Lag-year rather than current-year weights	−0.975***	0.272	0.0007	27,752
Unweighted	−0.806***	0.242	0.0015	27,752
Twelve eligible months in both years	−0.862**	0.345	0.0152	20,522
Quadratic zero-incidence control	−0.982***	0.273	0.0007	27,752
Quadratic and cubic zero-incidence controls	−0.901***	0.298	0.0037	27,752

Coefficients and standard errors are percentage points per one early-CPS incremental-signal standard deviation. The aggregate regression uses 300 lag-cell-by-outcome-year observations and signal-cell-clustered inference with 59 reference degrees of freedom. Other rows retain the primary person-transition frame except where the row changes weights or monthly coverage. Superscripts use unadjusted two-sided conditional tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. These diagnostics were conducted post hoc; their p -values are not multiplicity adjusted and omit CPS first-stage uncertainty. The nonlinear-zero rows address misspecification of the outcome equation's zero-incidence control without redefining the first-stage signal.

Leaving out each of the 60 signal cells produces entry coefficients from −1.223 to −0.890 point; all are negative and the largest unadjusted conditional p -value is 0.0020. Leaving out each of the 11 represented broad industries gives a range from −1.316 to −0.700 point, again all negative, with maximum $p = 0.0155$. These omission grids are influence summaries, not 60- or 11-test confirmatory families.

D Identification and Construct Checks for the Entry Result

This appendix reports the checks summarized in Section 4.3. Every regression uses the locked strict-post entry risk set, controls, fixed effects, current SIPP weights, and two-way clustering by person and lagged CPS cell unless a row states otherwise, and none propagates CPS first-stage uncertainty.

Table D.1. Within-person changes in the assigned tails

Design	Lower L	Upper U	ΔL	ΔU	Obs.	Persons
Table 1 baseline (lagged cell)	−2.712 (0.646)	1.917 (0.846)			27,752	19,300
Lagged levels plus changes, full frame	−2.623 (0.664)	1.988 (0.847)	0.058 (0.408)	0.315 (0.397)	27,752	19,300
Movers only, changes given lagged levels	−1.669 (1.242)	5.347 (1.163)	−0.057 (0.567)	0.563 (0.546)	6,231	5,338
Current-cell assignment, full frame	−0.416 (0.440)	0.609 (0.413)			27,752	19,300
Current-cell assignment, movers only	−0.042 (0.567)	0.647 (0.547)			6,231	5,338
Person fixed effects, lagged cell	−0.578 (1.303)	4.266 (1.987)			14,241	5,789
Person fixed effects, current cell	−0.046 (0.964)	−1.171 (0.752)			14,241	5,789

Coefficients are percentage points per one cross-cell standard deviation of the named early-CPS component; standard errors in parentheses. A mover is a transition whose signal cell differs between the lagged and current observation (6,231 transitions, 22.5 percent); ΔL and ΔU are the current minus lagged standardized distances and equal zero for stayers. The full-frame change specification also includes a mover indicator, whose coefficient is 2.057 (SE 0.439). Person-fixed-effect rows use the 5,789 workers with at least two entry-risk transitions, 2,226 of whom change cell, and replace the year and lagged-cell fixed effects by person and year effects. Among movers the standard deviations of ΔL and ΔU are 1.21 and 1.27 and their correlation is 0.56.

Table D.2. Hold-out year and split-sample heterogeneity

Sample	Lower L	Upper U	Obs.	Entry rate
<i>Panel A. Outcome years</i>				
Locked frame, outcomes 2020–2024	–2.712 (0.646)	1.917 (0.846)	27,752	0.088
Hold-out, 2018 to 2019 transitions	–0.343 (1.079)	–0.462 (0.987)	6,033	0.066
Pooled, outcomes 2019–2024	–2.419 (0.593)	1.573 (0.738)	33,785	0.085
<i>Panel B. Lagged bank-account balances, locked frame</i>				
Below \$1,000	–0.526 (0.754)	1.981 (0.877)	9,759	0.045
\$1,000 to \$7,500	–1.012 (0.948)	1.565 (0.983)	8,903	0.077
Above \$7,500	–6.204 (1.681)	0.643 (1.782)	9,090	0.143
<i>Panel C. Retirement account at $t - 1$, locked frame</i>				
None	–1.892 (0.610)	0.031 (0.699)	14,036	0.061
Owns	–3.108 (1.232)	3.566 (1.121)	13,716	0.115
<i>Panel D. Age band, locked frame</i>				
25–44	0.950 (1.249)	3.523 (1.785)	13,765	0.091
45–65	–2.885 (0.698)	3.593 (2.095)	13,987	0.084

Coefficients are percentage points per one cross-cell standard deviation; standard errors in parentheses. The hold-out row uses consecutive 2018 to 2019 worker-years in the same 60 supported cells with the same lagged assignment, controls, and clustering; the strict design excludes them because CPS destination earnings extend through 2018. Bank-balance terciles use weighted cutpoints from the locked frame and lagged annual means of the public SIPP monthly balance. Joint interaction tests that hold controls and fixed effects common across bank-balance terciles do not reject equal lower-tail or upper-tail coefficients within the locked frame ($p = 0.385$ and 0.890), within the 2018–2019 hold-out frame ($p = 0.661$ and 0.665), or within the pooled 2019–2024 outcome frame ($p = 0.373$ and 0.862), respectively. A continuous interaction with the inverse hyperbolic sine of the balance gives -0.154 (SE 0.080) for the lower and -0.089 (SE 0.097) for the upper tail. Adding the worker's own realized arc growth over the transition year does not move the upper coefficient in the bottom tercile (1.981 to 1.978).

Table D.3. Assigned historical CPS tails and individual SCE earnings expectations

Perceived object (SCE respondent-month)	Assigned L	Assigned U	Joint p	Obs.
Lower quantile distance, median minus 25th	0.122 (0.097)	–0.137 (0.062)	0.018	75,669
Upper quantile distance, 75th minus median	0.094 (0.088)	–0.154 (0.064)	0.016	75,669
Probability of a fall of more than 4%	0.0070 (0.0051)	–0.0080 (0.0031)	0.016	75,863
Probability of a rise of more than 4%	–0.0093 (0.0146)	–0.0065 (0.0079)	0.635	75,863
Perceived standard deviation	0.149 (0.118)	–0.176 (0.074)	0.015	75,863
Job-loss probability, next 12 months	0.070 (0.853)	–0.918 (0.516)	0.115	68,813

Assigned tails are CPS early-window (origins through 2017) lower and upper distances in 144 state-by-education-by-age cells that clear the Kish support rule of 200, out of 306 candidates, standardized across supported cells and matched to 75,863 SCE core responses from 10,861 respondents aged under 60 in 143 cells. Each row regresses the perceived object on the two assigned tails, the assigned zero share, quadratic mean and dispersion, household-income and numeracy categories, and state, education, age-category, and year fixed effects, weighted by the SCE sampling weight with standard errors clustered by cell. Perceived quantile distances and the standard deviation are in percentage points of expected earnings growth; probabilities are shares. In outcome standard deviations the largest coefficient, that of the assigned upper tail on the upper quantile distance, is -0.07 .

Table D.4. Finer assignment cells: occupation groups

Cell scheme	Cells	Raw rel.	Incr. rel.	Lower L	Upper U	Incr. Kelley	Obs.
Industry, education, age (paper)	60	0.710	0.390	−2.712 (0.646)	1.917 (0.846)	−1.047 (0.272)	27,752
Industry, occupation, education	69	0.671	0.315	−0.666 (0.725)	0.316 (0.860)	−0.369 (0.227)	26,290
Industry, occupation, education, age	96	0.529	0.180	−1.218 (0.976)	0.552 (0.906)	−0.438 (0.286)	23,735
Occupation, education, age	18	0.817	0.344	−1.259 (0.531)	−1.738 (0.976)	−0.269 (0.134)	27,751

Occupation groups are professional and managerial (CPS OCC1990 3–199; SIPP 2018 codes 10–3655), sales, office, and administrative support (203–389; 4700–5940), and service, production, construction, and transportation (403–889; 3700–4655 and 6005–9760). SIPP workers receive their modal annual occupation group. Cells are early-CPS cells clearing Kish support of 200. Reliability is the Spearman–Brown transform of the mean correlation across 30 random global person splits generated inside the script, which is why the paper’s scheme shows 0.710 and 0.390 rather than the 0.687 and 0.397 of the frozen deterministic splits in Table 5. Coefficients are percentage points per one cross-cell standard deviation under the scheme’s own standardization, with lagged assignment, the scheme’s zero-share, mean, and dispersion controls, the paper’s household controls and fixed effects, and clustering by person and lagged cell. Observations fall when a worker-year lacks a supported cell under the finer scheme.

E Portfolio Parameterizations

Table E.1. Broader 2018–2024 early-CPS to later-SIPP specification path

Outcome	Raw	Raw + controls	Incremental	BH q
Participation	0.004 (0.007)	−0.022* (0.012)	−0.008* (0.005)	0.106
Share, positive financial assets	0.097 (0.162)	−0.552** (0.219)	−0.209** (0.083)	0.058
Share, all workers	0.115 (0.163)	−0.489** (0.227)	−0.185** (0.086)	0.071
Share, current holders	0.200 (0.590)	0.070 (0.993)	0.027 (0.376)	0.944

This table is the broader 2018–2024 timing-sensitivity window, not the strict-post primary sample. Conditional two-way-clustered standard errors are in parentheses. Superscripts denote unadjusted two-sided conditional significance: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$; they omit BH adjustment and CPS first-stage uncertainty. Raw Kelley is standardized using early supported cells. “Raw plus controls” adds the linear CPS zero share and quadratic mean and dispersion; Incremental Kelley is the standardized residual from that projection. Those two columns are FWL-equivalent—their t statistics, p -values, and four-outcome q -values agree—and are not independent tests. Plug-in first-stage-uncertainty p -values are 0.182 for participation, 0.115 for the positive-assets share, and 0.168 for the all-worker share.

In this broader person-year window, the raw estimates are positive and imprecise. The conditional comparison changes sign for the broad margins. This sign change is the downstream consequence of changing the estimand; the person-year q -values do not pass a 5% family-adjusted threshold, and the CPS first-stage resampling sensitivity widens inference further. The retained broader-window signal-cell-year aggregation gives an incremental positive-assets-share coefficient of −0.224 (SE 0.085; 420 cell-years in 60 signal cells; four-outcome BH $q = 0.044$). The corresponding participation, all-worker-share, and holder-share coefficients are −0.007, −0.170, and 0.092, with q -values 0.198, 0.125, and 0.816. This isolated conditional result predates the strict timing restriction and does not incorporate CPS first-stage uncertainty, so it is not promoted to primary evidence. Sorting, historical rather than perceived risk, and incomplete portfolio measurement preclude a behavioral interpretation.

F Subjective-Belief Construct and Timing Audits

Several distinct literatures discipline this audit. Guiso, Jappelli, and Terlizzese (1996) relates directly elicited income uncertainty to household portfolios. Wang (2023), Caplin et al. (2026), and Arellano et al. (2026) show why perceived earnings risk cannot be substituted mechanically with risk estimated from realized panels. Drerup, Wibral, and Zimpelmann (2023) finds that allocations respond to elicited asset-return skewness conditional on other expected moments, while Giglio et al. (2021) shows that beliefs can predict adjustment direction conditional on trading without predicting whether an investor trades. Dew-Becker (2024) illustrates a useful construct discipline in a different setting: forward-looking and realized skewness are distinct objects, and relative upper and lower semivariance can separate orientation from total scale. Very close concurrent work by van den Akker et al. (2026) reports that a longer perceived earnings right tail predicts administrative stock and bond investment. None of those results validates the fixed-threshold earnings imbalance used here: the risk object, centering, portfolio outcome, and timing differ.

F.1 Fixed thresholds are not centered shape

Lemma 3 (Translation of a fixed-threshold imbalance). *Let X have distribution function F , with density f positive at the relevant thresholds, and let $c > 0$. For the translated variable $X + m$, define*

$$A_c(m) = \Pr(X + m < -c) - \Pr(X + m > c) = F(-c - m) + F(c - m) - 1.$$

Then $A'_c(m) = -f(-c - m) - f(c - m) < 0$, although every translation-invariant centered shape measure is unchanged.

Proof. Continuity gives $\Pr(X + m < -c) = F(-c - m)$ and $\Pr(X + m > c) = 1 - F(c - m)$. Subtraction yields the expression; differentiation yields the derivative. A translation-invariant centered-shape functional takes the same value for X and $X + m$. $\square$

The public SCE density bins use fixed percentage-growth thresholds. The imbalance subtracts probability above +4 percent from probability below −4 percent. In the state-education-year reconstruction, this threshold imbalance is correlated −0.941 with the forecast mean. Additive cubics in the mean and standard deviation explain 89.4 percent of its variation, leaving 32.6 percent of its raw standard deviation. The residual correlates 0.018 with centered Bowley shape and 0.029 with centered moment skewness. Residualizing a fixed-threshold statistic therefore does not recover the same ranking as a centered shape estimator.

F.2 Prospective public SCE-to-SIPP exercise

SCE beliefs are aggregated by state, education, and year and lagged before a subsequent SIPP portfolio change. The prospective merge contains 42,375 consecutive SIPP transitions on 26,005 people, spanning 31 states and 366 state-education-outcome-year cells. The specification absorbs

state-by-year, state-by-education, and education-by-year effects, adds perceived mean and dispersion and baseline household controls, and clusters by state. It is a construct-and-timing comparison, not causal evidence.

Table F.1. Prospective SCE-to-SIPP construct comparison

Belief signal for subsequent active share change	Coefficient	S.E.	<i>p</i> -value
Fixed-threshold residual	0.1543	0.2136	0.476
Centered Bowley downside shape	−0.3471*	0.1997	0.093
Centered moment downside skewness	−0.0121	0.1745	0.945

Standard errors are clustered by state; superscripts denote unadjusted two-sided significance: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. Split reliability is 0.198 for the raw threshold, 0.147 for its leave-year-out residual, 0.068 for Bowley shape, and 0.121 for moment skewness. The centered estimators do not agree, and the threshold result is null.

F.3 Household Finance timing reconstruction

The Household Finance action is the reported direction of retirement reallocation in the defined-contribution account over the preceding twelve months, coded +1 for a move into riskier assets, −1 for a move into safer assets, and 0 for no change; every belief regressor keeps the paper’s downside-positive orientation, so a positive coefficient means that perceived downside predicts moving toward risk. The action window is retrospective. A same-calendar-year average of recurring belief reports is not prospectively ordered: among respondents with a coded reallocation answer and at least one same-calendar-year belief report, 87.1 percent have at least one report recorded after the module month, and no same-calendar-year report can predate the start of the twelve-month action window. The reproducible reconstruction therefore uses only reports one through eleven months before the module, the official module weight, and forecast mean and standard deviation controls.

Table F.2. Timing-ordered Household Finance reconstruction

Belief window for the reallocation outcome	Fixed-threshold imbalance	Centered Bowley downside shape	<i>N</i>
<i>Panel A. Timing grid (outcome: +1 riskier, −1 safer, 0 no change)</i>			
Pre-module: months −11 to −1	0.067** (0.031)	−0.024* (0.013)	2,136
no intervening realized shock	0.013 (0.051)	−0.037** (0.017)	1,270
realized large income fall only	0.240*** (0.083)	0.015 (0.035)	190
Post-module: months +1 to +11	−0.052 (0.041)	−0.010 (0.015)	1,805
Joint design: pre-module term	0.109** (0.049)	−0.022 (0.015)	1,749
Joint design: post-module term	−0.091* (0.047)	0.001 (0.017)	1,749
<i>Panel B. Standing DC equity share: perceived-dispersion coefficient</i>			
Perceived s.d. (fixed-threshold control)	−3.633*** (1.002)		3,332
Perceived s.d. (Bowley-shape control)	−3.766*** (1.006)		3,332
Perceived s.d. (moment-skewness control)	−4.734*** (0.961)		2,952

Standard errors, in parentheses, are clustered by state; superscripts denote unadjusted two-sided significance: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. This timing and subgroup grid is exploratory, and the reported stars and p -values are not adjusted for the family of comparisons in the table. Every specification uses the official Household Finance quarterly weight and Household Finance year and state fixed effects. Panel A additionally controls for forecast mean, perceived dispersion, age, education, and household income. Post-module beliefs cannot have caused the retrospective action, so the post-module row is a negative control; the joint design enters the pre-module and post-module signals together ($N = 1,749$). N refers to the fixed-threshold column; Bowley samples differ by at most one respondent. Centered moment skewness, run on the same grid, is small and unstable: coefficients range from -0.083 to $+0.026$, and the only entries with $p < 0.05$ are the realized-fall subsample and the joint post-module term. In Panel B, perceived dispersion is the focal regressor rather than a control; each row includes the named shape control plus forecast mean, age, education, and household income. Panel B uses the pre-module construction and reports percentage-point changes in the standing DC equity share per one-standard-deviation increase in perceived earnings dispersion. The equity share is an accumulated stock and beliefs are not assigned, so Panel B is a descriptive scale benchmark rather than a causal estimate.

The grid locates the fixed-threshold measure’s apparent pre-module association. Its point estimate is 0.013 when respondents with an intervening realized shock are excluded and 0.240 in the 190-person realized-large-fall subsample; this contrast is exploratory and is not a decomposition of the full-sample estimate. The post-module estimate is -0.052 , and post-module beliefs cannot have caused the retrospective action. Entered jointly, the pre-module and post-module terms carry opposite signs ($+0.109$ and -0.091). A belief window that mixes reports from both sides of the module—as 87.1 percent of same-calendar-year report sets do—therefore blends differently signed components and can return either sign. A realized income fall occurring after a pre-module belief report also cannot have imprinted itself on that earlier report; the subgroup contrast may instead reflect selection, common shocks, or the action window’s overlap with the intervening period. Combined with Lemma 3’s result that the fixed-threshold statistic is dominated by the forecast mean, the grid shows sensitivity to construct and timing choices, not a prospective orientation signal or an identified belief-formation mechanism.

Centered Bowley shape has the background-risk sign in the pre-module rows, a larger magnitude when realized shocks are excluded, and an estimate near zero in the post-module row. It is not corroborated by moment skewness, the comparisons are exploratory, and its magnitude is modest, so

it is reported as a directional diagnostic rather than as validation or evidence about how beliefs are formed. The dispersion benchmark, by contrast, is stable across all three shape controls. Together the exercises show that the linked survey data contain economically relevant scale information while reinforcing the paper’s narrower lesson: temporal ordering, centered construction, independent reliability, and outcome inference are separate gates.

F.4 Centered upper and lower semimoments

The fixed-threshold statistic does not isolate tail orientation, so I reconstruct each respondent-month’s ten-bin public density. Open tails are assigned ± 16 percentage points, the three reports whose integer bins sum to 99 or 101 are renormalized, and lower and upper semideviations are centered on the respondent’s own elicited mean. Let d_i^- and d_i^+ denote those semideviations. The focal regression enters their sum and the tilt $d_i^+ - d_i^-$; a positive tilt coefficient means that shifting perceived tail magnitude from downside toward upside is associated with a riskier reported reallocation. This is an absolute semideviation-gap contrast, not a scale-free orientation measure.

Table F.3. Perceived earnings-tail tilt and reported retirement reallocation

Specification or outcome	Coef.	S.E.	p-value	BH q	N
Pre-module reallocation: primary tilt	0.0276**	0.0128	0.036	0.054	2,136
equal weight for each belief month	0.0284**	0.0127	0.030	—	2,136
no reported large change or contribution cut	0.0435***	0.0137	0.003	0.008	1,270
Post-module reallocation: negative control	0.0047	0.0178	0.793	—	1,805
Pre-module reallocation: normalized semivariance orientation	0.0189	0.0137	0.173	0.347	1,893
Standing DC equity share: primary tilt	−0.576	0.664	0.390	0.390	3,332

The reallocation outcome equals +1 for a reported move into riskier assets, −1 for a move into safer assets, and zero for no change over the preceding twelve months. Except for the explicitly equal-month row, repeated pre-module belief reports are averaged within respondent using the public monthly SCE sampling weight. The primary tilt is the standardized upside-minus-downside semideviation gap entered jointly with total semideviation scale, expected earnings growth, age, education, household income, Household Finance year fixed effects, and state fixed effects. The normalized row divides the upper-minus-lower semivariance difference by total variance and controls for total standard deviation. Official Household Finance quarterly weights are used; standard errors are clustered by state. Superscripts denote unadjusted two-sided significance: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

The primary $q = 0.054$ adjusts the three displayed focal shape tests in its outcome family; expanding to fourteen support and outcome variants gives $q = 0.169$. The no-large-change restriction excludes a reported large income, health, or wealth change and a respondent or spouse retirement-contribution cut; it is a post hoc subgroup and its stronger coefficient is scale dependent. The post-module report cannot have caused the retrospective action, but the formal pre-minus-post contrast is also imprecise ($p = 0.191$). Beliefs recorded before the module usually fall inside the retrospective action window. The separate lower and upper semideviations are correlated 0.95 and are therefore not presented as independent discoveries. The result is exploratory and descriptive: it does not survive the full support family, normalized orientation is null, standing share is null, and adding the tilt changes five-fold cross-validated RMSE by only −0.065%.

The main gap coefficient is positive because the separate components carry the economically coherent signs: lower semideviation is −0.107 (SE 0.047; $p = 0.026$) and upper semideviation is +0.087 (SE 0.045; $p = 0.060$) when entered jointly. Their 0.95 correlation makes the gap-plus-sum rotation the more transparent parameterization. In the full sample, the tilt mainly predicts a lower probability of reporting a move into safer assets rather than a statistically precise increase in moves into riskier assets; among active movers alone it is null. Thus the individual SCE densities supply a

directional perceived-upside pattern worth reporting, but not evidence that perceived risk dominates assigned histories and not a second headline result.

G Plug-In First-Stage Uncertainty

Suppose the second-stage regression assigns $\hat{K}_c = T(\hat{G}_c)$. A conventional variance conditions on the observed $\hat{K}_c$. Clustering by c permits shared second-stage residuals but does not integrate over repeated samples of the nonlinear first-stage functional.

The CPS bootstrap resamples unique people within each frozen signal cell. If a draw repeats a person m times, weighted first moments use mw and Kish denominators use mw^2 , not $(mw)^2$. Every draw rebuilds weighted mean, standard deviation, zero share, quantiles, support selection, equation (5), standardization, SIPP assignment, and the outcome regression. Across 200 draws, all requested estimates succeed; early support ranges from 59 to 61 cells and pooled support from 63 to 66. The strict-post level coefficient draws are not perfectly centered on the corrected point estimates. For participation, their mean is -0.0062 versus a point estimate of -0.0092 (mean-minus-point 0.0029 ; 2.5th–97.5th percentiles $[-0.0145, 0.0013]$). For the all-worker share, the corresponding values are -0.150 versus -0.234 (mean-minus-point 0.084 ; percentiles $[-0.363, 0.045]$).

The transition bootstrap additionally reassigns both annual observations and rebuilds the risk sets in each draw. For nonholder-to-positive-holdings entry, the draw mean is -0.650 point versus a point estimate of -1.047 points, a mean-minus-point displacement of 0.397 point. The across-draw standard deviation is 0.258 point, 198 of 200 coefficients are negative, and the 2.5th–97.5th percentile range is $[-1.125, -0.138]$ points. For signed change the corresponding standard deviation is 0.250 point; for exit it is 0.845 point. These ranges are descriptive summaries of rebuilt-CPS coefficients with SIPP held fixed, not confidence intervals, probability statements, or bias corrections; their displacement further motivates treating the variance combination as a sensitivity.

Because CPS and SIPP are independent, the paper reports the following plug-in first-stage uncertainty calculation:

$$\widehat{\text{Var}}_{\text{plug-in}}(\hat{\beta}) = \widehat{\text{Var}}_{\text{SIPP}}(\hat{\beta} \mid \hat{G}) + \widehat{\text{Var}}_{\text{CPS}}(\hat{\beta}).$$

The first term is the conditional two-way-clustered SIPP variance evaluated at the observed signal, and the second is the across-draw variance induced by resampling CPS people and rerunning the stated first-stage pipeline. This is not a joint nonparametric bootstrap of CPS and SIPP. It propagates CPS signal-estimation uncertainty within the frozen cell schema, crosswalk, and observed SIPP sample; it does not cover crosswalk or functional-form choice, SIPP sample-reconstruction uncertainty, or a latent-shape correction. The resampler also does not model cross-cell covariance from a person appearing in multiple cells, but no CPS person does so in the frozen annual-link sample. Reported plug-in p -values use a two-sided standard-normal reference. The primary strict-post first-stage results use this calculation; Table G.1 reports the corresponding design sensitivity.

Table G.1. Strict-post power for the early incremental signal

Outcome	80% minimum detectable effect		Power at observed magnitude	
	Conditional	CPS plug-in	Conditional	CPS plug-in
Participation	1.34	1.79	0.481	0.298
All-worker asset share	0.273	0.409	0.672	0.361

Entries use the strict-post 2019–2024 sample and are per one standard deviation of the early CPS incremental signal. Both MDE columns are in percentage points. The first pair gives two-sided 5%-size, 80%-power minimum detectable effects under a normal approximation; the second gives power at the absolute magnitude of the observed coefficient. “Conditional” treats the estimated CPS signal as fixed. “CPS plug-in” uses the conditional variance plus the CPS first-stage bootstrap variance and remains a sensitivity rather than joint CPS–SIPP inference. The table deliberately reports no latent-signal correction: weak cross-survey transport and possible nonclassical error preclude an equivalence mapping from the assigned signal to latent orientation.

Table G.2. Implementation checklist for group-estimated earnings-risk regressors

Layer	Required disclosure	Diagnostic reported here
Population and support	Define origin population, destination zeros, growth transformation, cell schema, weights, and effective support.	Positive-origin workers; zero-inclusive arc growth; exact crosswalk; Kish support.
Reproducibility	Split globally by person and report rank and time stability.	Person-split Spearman–Brown reliability and early-versus-late correlations.
Incremental content	State the conditional interpretation and report the sequential relation with zero share, mean, and dispersion.	Raw; after zero share; after zero share, mean, and dispersion.
Downstream inference	Preserve support, nonlinear estimation, projection, standardization, and assignment inside first-stage resampling.	Independent CPS person bootstrap combined with conditional SIPP variance.

H Scope of the Group-Generated-Risk Workflow

The protocol is motivated by a class of designs rather than by a claim that every paper below makes the same choices or faces the same data constraint. Table H.1 records representative finance applications in which a risk object is estimated from one history or grouping and then used in a portfolio, consumption, or performance exercise. Rich administrative designs can make some gates much less binding; the common feature is that construction of the estimated risk object remains part of the downstream estimand.

Table H.1. Illustrative workflows using estimated risk objects

Study	Estimated object	Downstream use	Relation to this protocol
Ranish (2013)	Permanent variance, disaster exposure, and cyclical labor-income risk estimated in PSID histories and mapped to SCF households	Household stockholding and leverage	Direct two-sample group assignment; bootstraps estimated risk, multiple imputation, and generated-regressor uncertainty
Angerer and Lam (2009)	Occupation-by-education permanent and transitory income risk	Household risky-asset participation and shares	Early public group-risk assignment; does not isolate a nonlinear tail-orientation contrast
Fagereng, Guiso, and Pistaferri (2018)	Firm-level shocks to individual wage income	Household risky shares	Administrative and instrumental-variables benchmark for the effect of wage risk rather than a public-cell measurement design
Chang et al. (2022)	Individual income-volatility breaks and firm-volatility instruments	Household portfolio choice	Causal second-moment benchmark; distinct from higher-moment reliability and transport
d'Astous and Shore (2024)	Permanent income volatility shifted by university-admission discontinuities	Household participation and risky shares	Quasi-experimental sorting benchmark; stronger identification but a different risk object
Shen (2024)	Regional earnings-risk skewness	PSID consumption and risky portfolio shares	Nonlinear group risk assigned to households; cell support and incremental shape are relevant interpretation gates
Catherine, Sodini, and Zhang (2024)	Education-by-industry cyclical income skewness in Swedish registers	Household equity participation and shares	High-information administrative benchmark for the same risk-to-portfolio logic
Catherine (2022)	Estimated non-Gaussian, state-contingent income process	Life-cycle portfolio model	Shows the economic object that a reduced-form group statistic seeks to proxy
Gálvez and Paz-Pardo (2026)	Nonlinear earnings process estimated from PSID histories	Structurally estimated consumption, housing, participation, and risky shares	Substantive benchmark showing that rich nonlinear risk can matter; it does not test a public group-assigned shape statistic
Akey, Robertson, and Simutin (2026)	Methodologically revised Fama–French factor vintages	Mutual-fund performance, stock alphas and betas, and corporate-finance conclusions	Finance analogue showing that construction revisions can change downstream inference

Notes: The table establishes empirical scope; it is not an allegation that the cited papers fail a universal checklist. Their data, estimands, and identification strategies differ. The present contribution isolates three disclosures that are especially important when nonlinear public-cell statistics are assigned across samples: reproducibility, incremental content, and first-stage uncertainty accounting.